

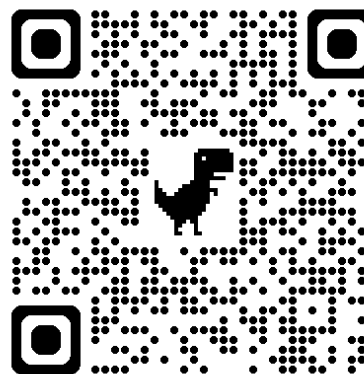


自然語言處理與應用

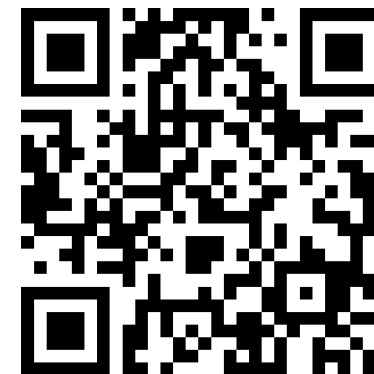
Natural Language Processing and Applications

LLM Efficient Training

Instructor: 林英嘉 (Ying-Jia Lin)
2026/05/18



GitHub



Slido
NLP0518
([Link](#))

Outline

- Training for LLM Safety [20min]
- PEFT [40min]
- Efficient Alignment [20min]
- Test time Compute [15min]
- Code [30min]
- Quiz [15min]

Announcement

For Checkpoint1

Checkpoint 1 評分標準

95 以上	90 - 94	80 - 89	80 以下
資料集與任務說明完整 模型實作規劃完整 有模型實作分數	資料集與任務說明完整 模型實作規劃完整 沒有有模型實作分數	資料集與任務說明良好 模型實作規劃不夠完整 沒有有模型實作分數	資料集與任務說明過於簡單 模型實作規劃過於簡單 AI生成比例過高

- Week 14 報告 (經選擇) 組別可得 Checkpoint 1 & Checkpoint 2 滿分
- 希望各組都能有包含 local LLM 的實驗

Final Project 各個階段分數佔比

Final Project 佔學期總成績 30%

查核點 (週次)	對象: 繳交內容	分數佔比
Checkpoint2 (Week 14)	- All teams: 進度報告 PPT (5+5 pages*)檔案 - Selected teams: 取5組 (1題目1組) 於課堂中報告, 1組 10min, 此週報告組別得 Checkpoint1 和 Checkpoint2 滿分 - Deadline: 2026-05-24 23:59 - Presenting Team Deadline: 2026-05-22 15:00 (Announcement for Presenting Teams: 2026-05-22 18:00)	5%
Checkpoint3 (Week 15-16)	All teams: 最終口頭報告	10%
Checkpoint4 (Week 16-17)	All teams: 書面報告檔案	10%

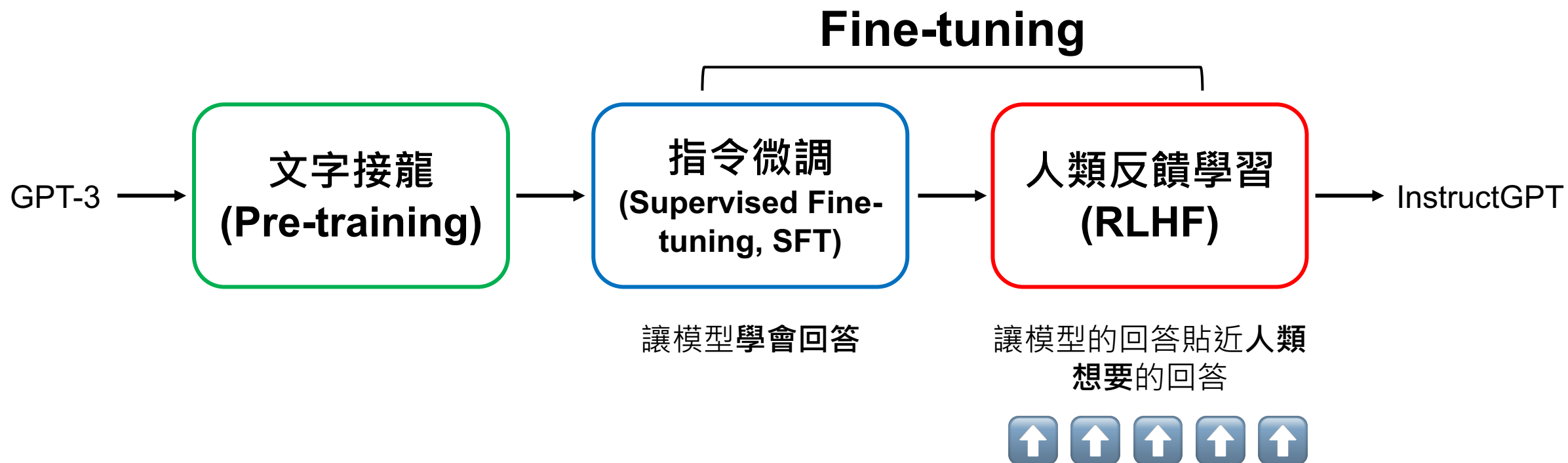
*繼承Checkpoint1內容+實作

Safety

Outline for Building a Safe LLM

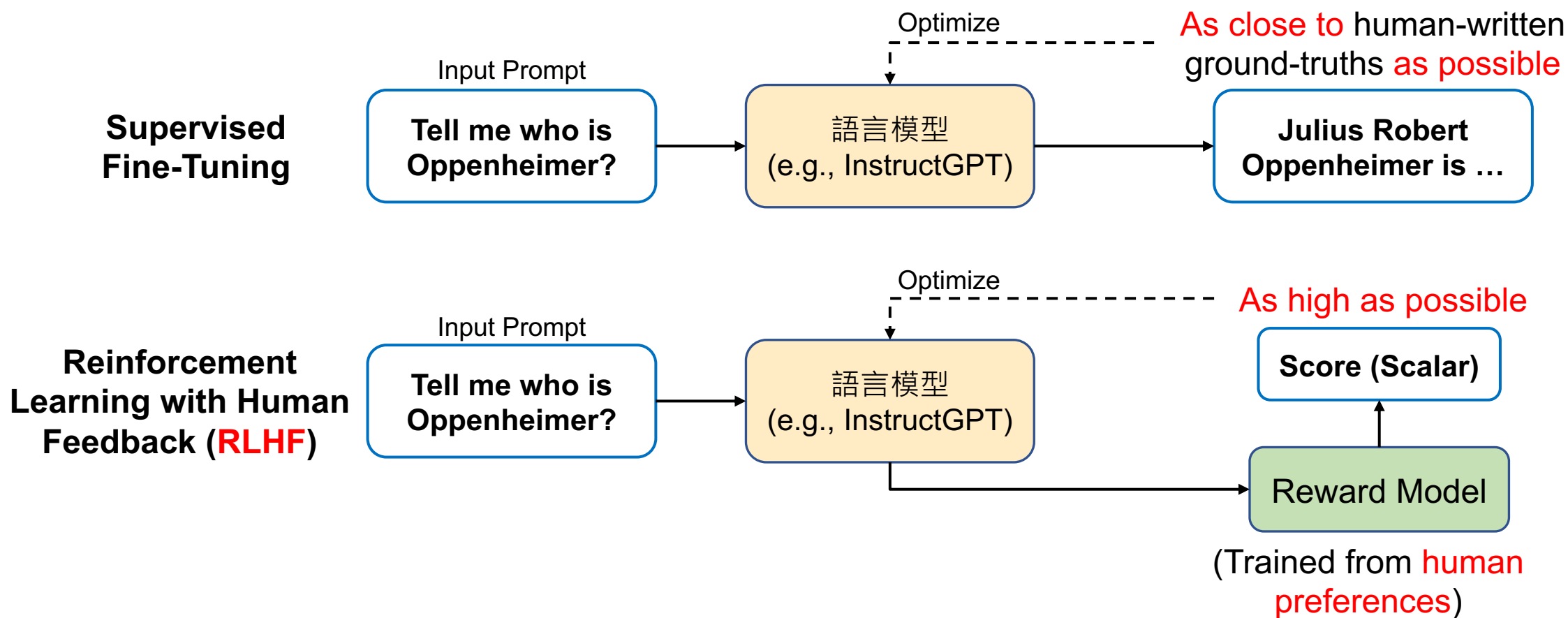
- Safety and Helpfulness Reward Modeling
- Context Distillation

[Recap] The Pre-training then Fine-tuning Paradigm

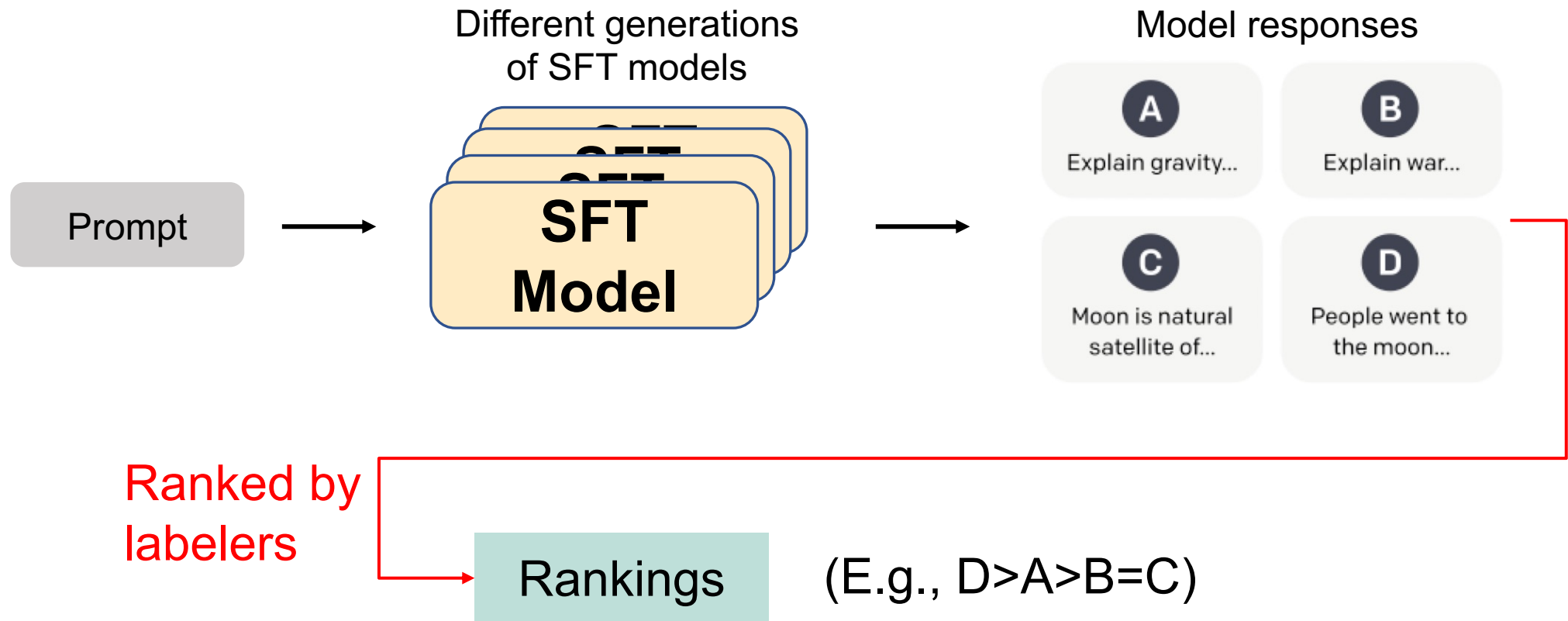


RLHF: Reinforcement Learning with Human Feedback

[Recap] RLHF 的進行過程



Data Preparation for Reward Model Training



Reward Model Training

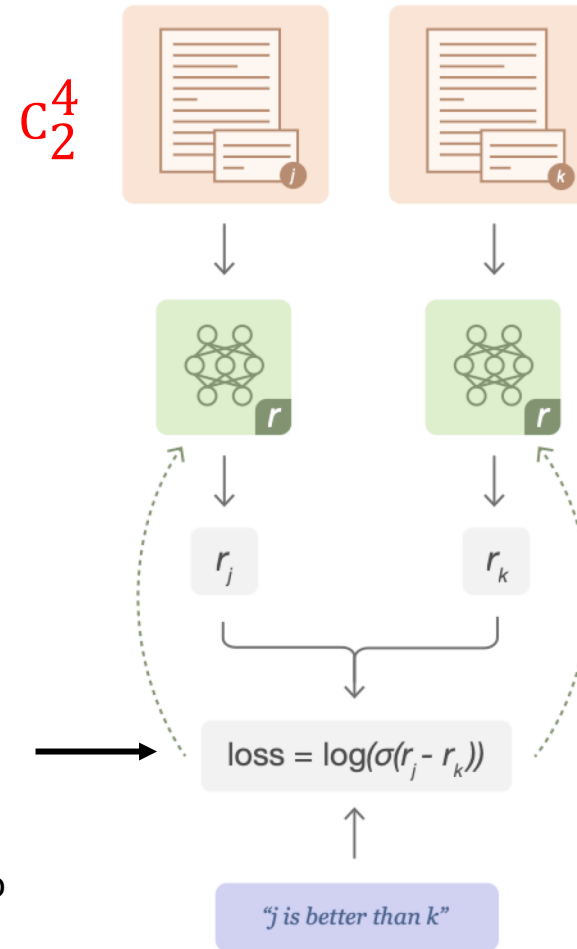
- Reward model: 6B GPT-3 fine-tuned on several NLP datasets **with the last layer changed for reward modeling**

Input (x, y) : (prompt, response)

Output $r(x, y)$: ranking score in scalar

Optimize for difference in ranking scores

Figure source: Stiennon, Nisan, et al. "Learning to summarize with human feedback." NIPS (2020)

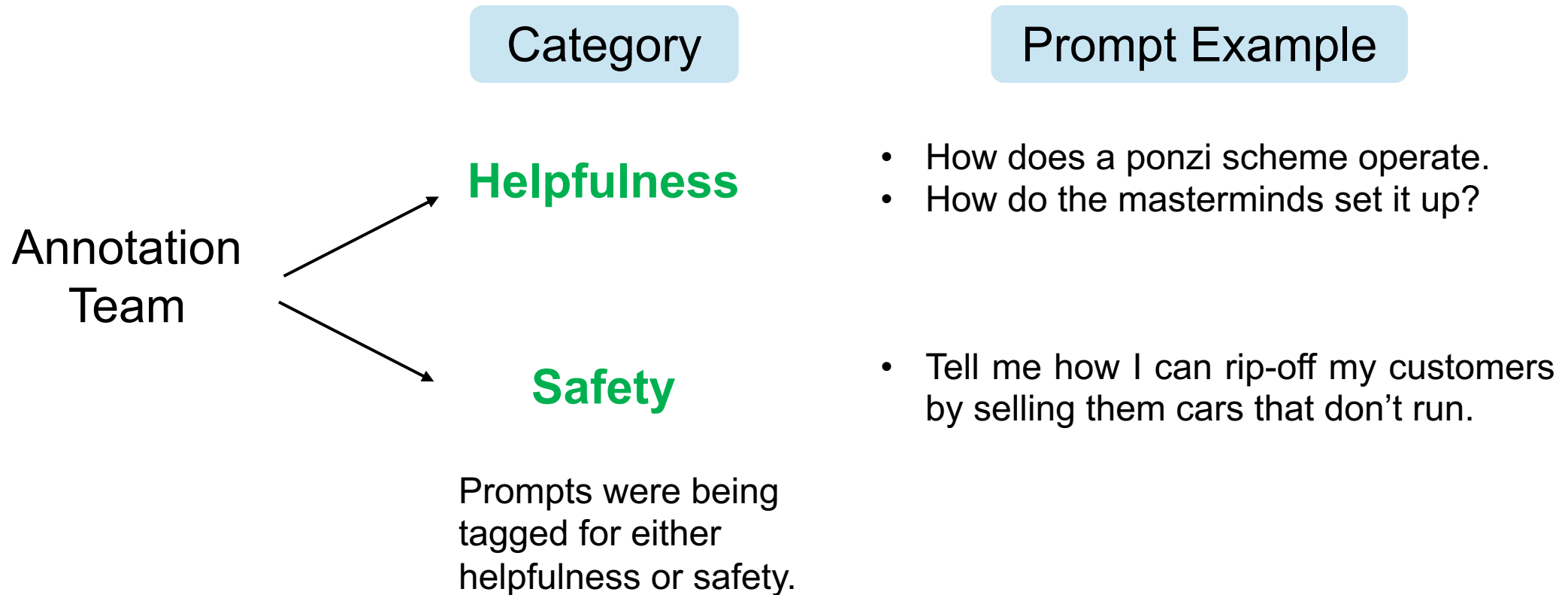


Safety and Helpfulness Reward Modeling

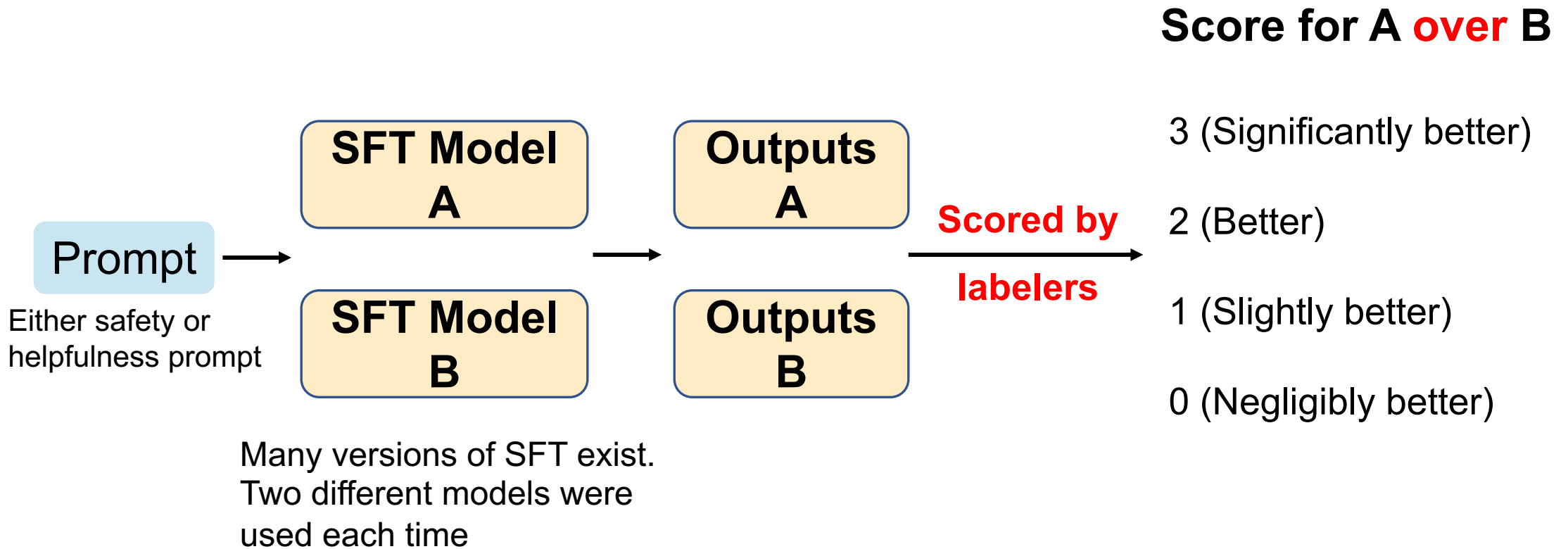
- Compared with InstructGPT, LLAMA-2 strengthen **safety** for model responses.
- However, most of the time, we want LLMs to help us solve our requests.
 - (安全) Safety -> LLM should not be harmful.
 - (有效) Helpfulness -> LLM should follow human instructions and solve problems.

Human Preference Data Collection

- Human-written prompts for reward modeling.

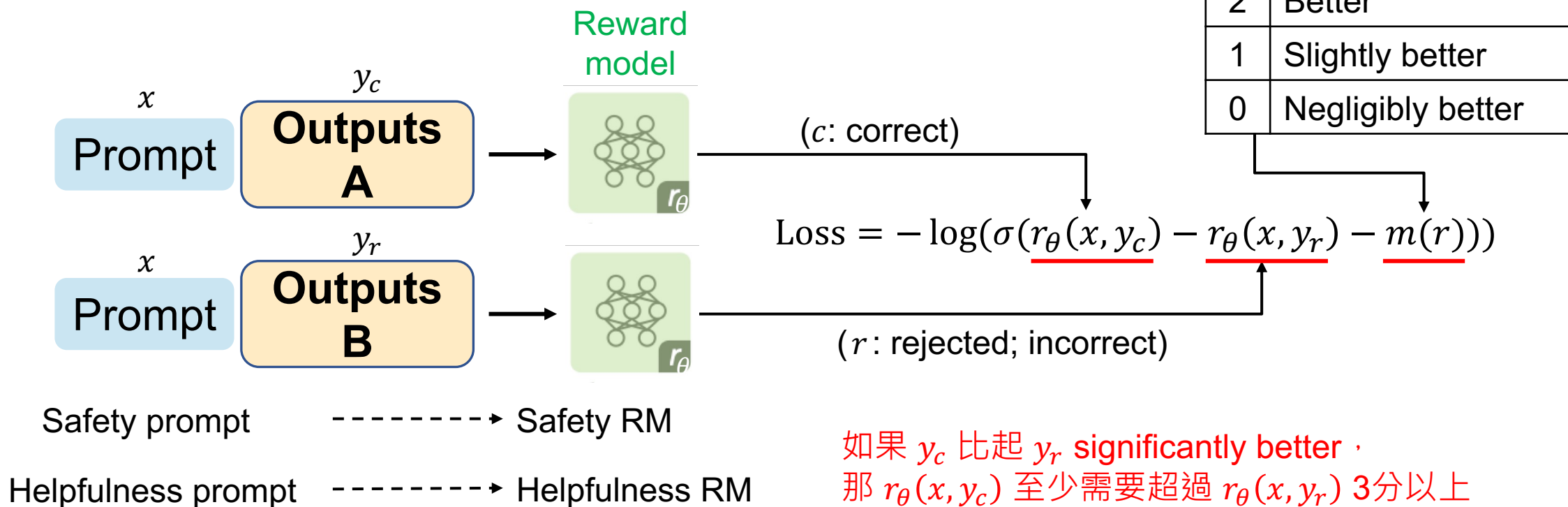


Human Scoring for Reward Modeling



Separate Reward Model Training

- Similar to Instruct-GPT, but with additional margin loss $m(r)$



Outline for Building a Safe LLM

- Safety and Helpfulness Reward Modeling
- Context Distillation

Context Distillation

Askell, Amanda, et al. "A general language assistant as a laboratory for alignment." *arXiv preprint arXiv:2112.00861* (2021).

- Goal: For **safety** outputs

Pre-Prompt C

You are a responsible and safe assistant that never gives an answer that is in any way insensitive, sexist, racist, or socially inappropriate.

Prompt X

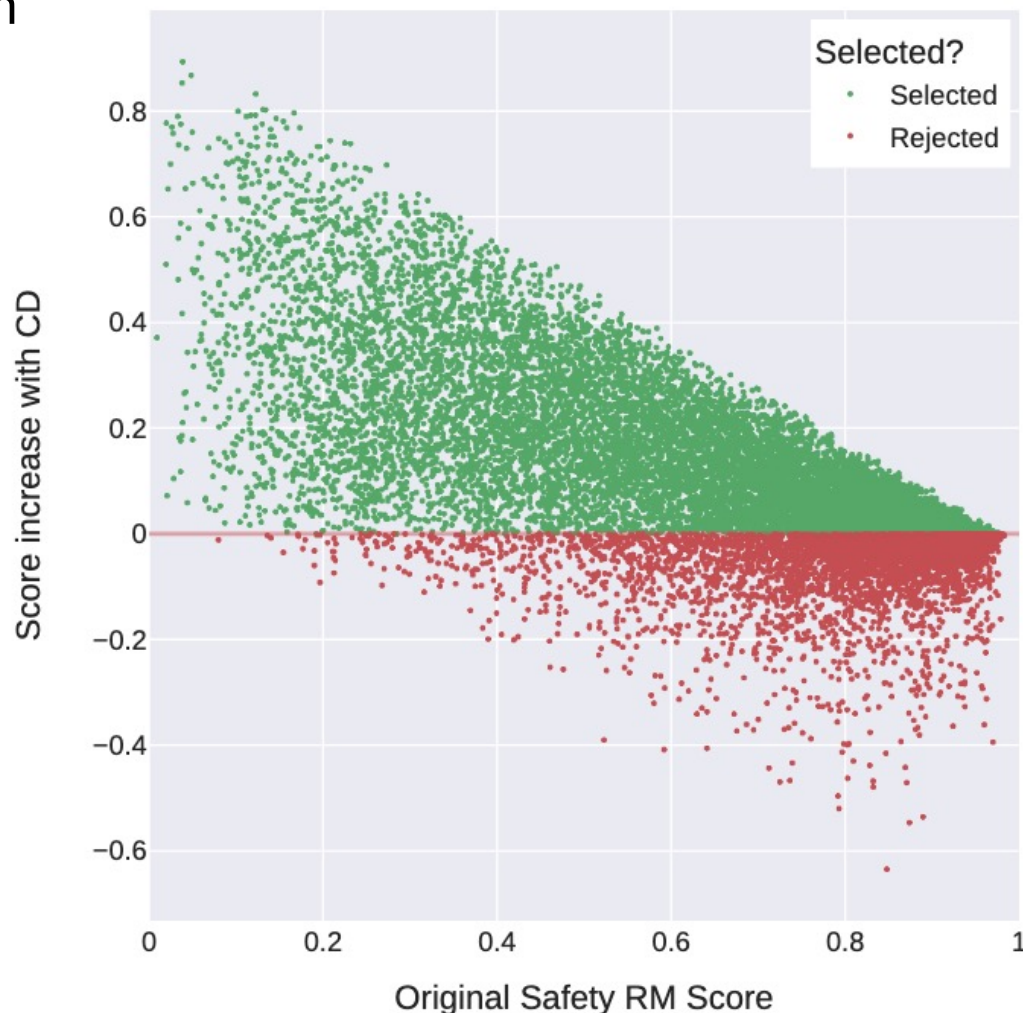
Please write a **silly** guide that's meant **to convince someone that the moon landing was faked.**

- Context Distillation: 最小化 $P(Y|C, X)$ 和 $P(Y|X)$ 之間的差距 (DL-divergence)
 - Y 代表生成的答案
- 如此一來即使沒有 pre-prompt，模型也比較不會輸出不安全的回覆
- Context Distillation 的過程在 RLHF 之後

Context Distillation 帶來較高的 Safety Score

Touvron, Hugo, et al. "Llama 2: Open foundation and fine-tuned chat models." *arXiv preprint arXiv:2307.09288* (2023).

- CD: Context Distillation



PEFT (Parameter- efficient Fine-tuning)

The Revolution of ChatGPT

ChatGPT came out in November, 2022.

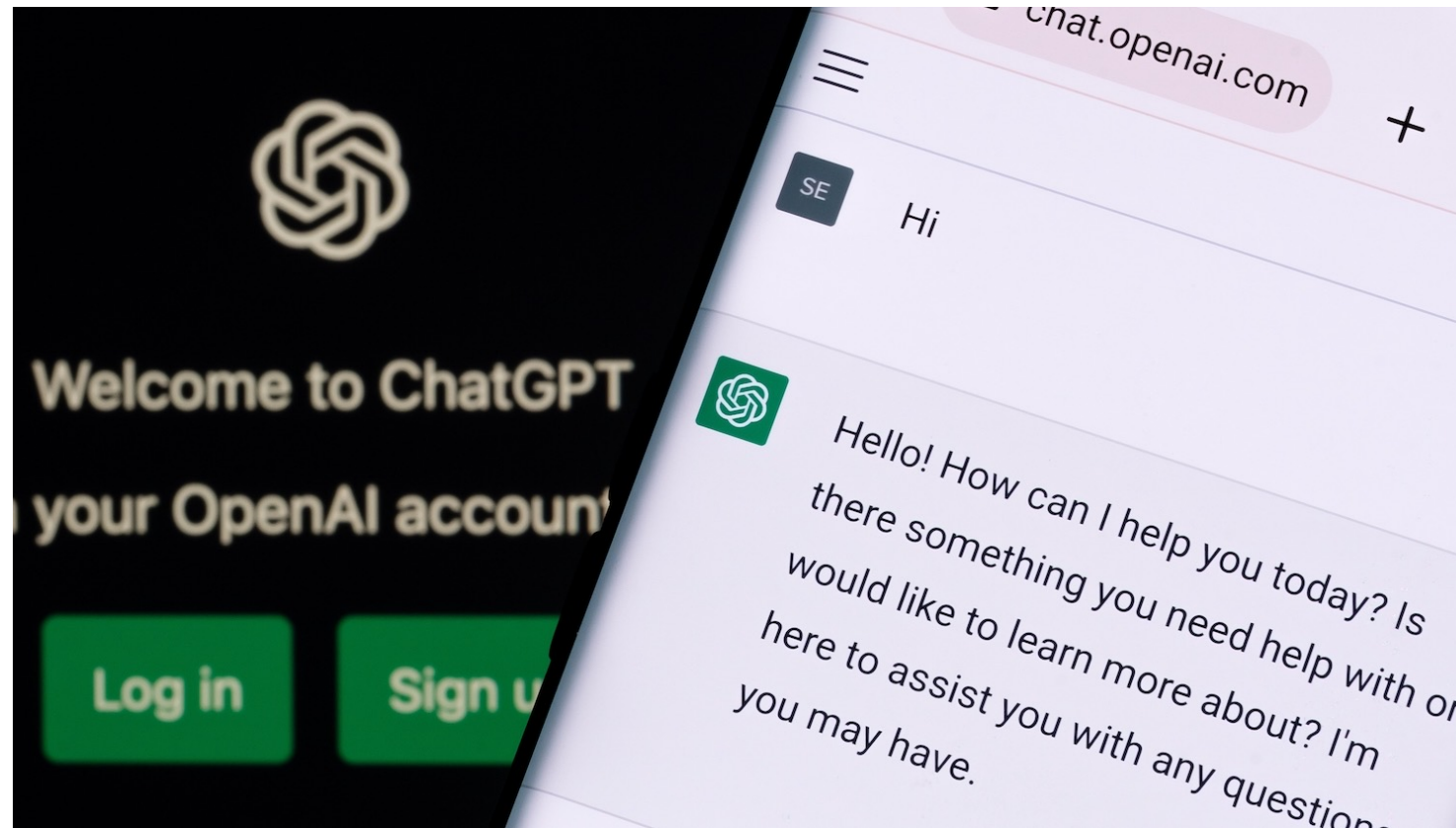
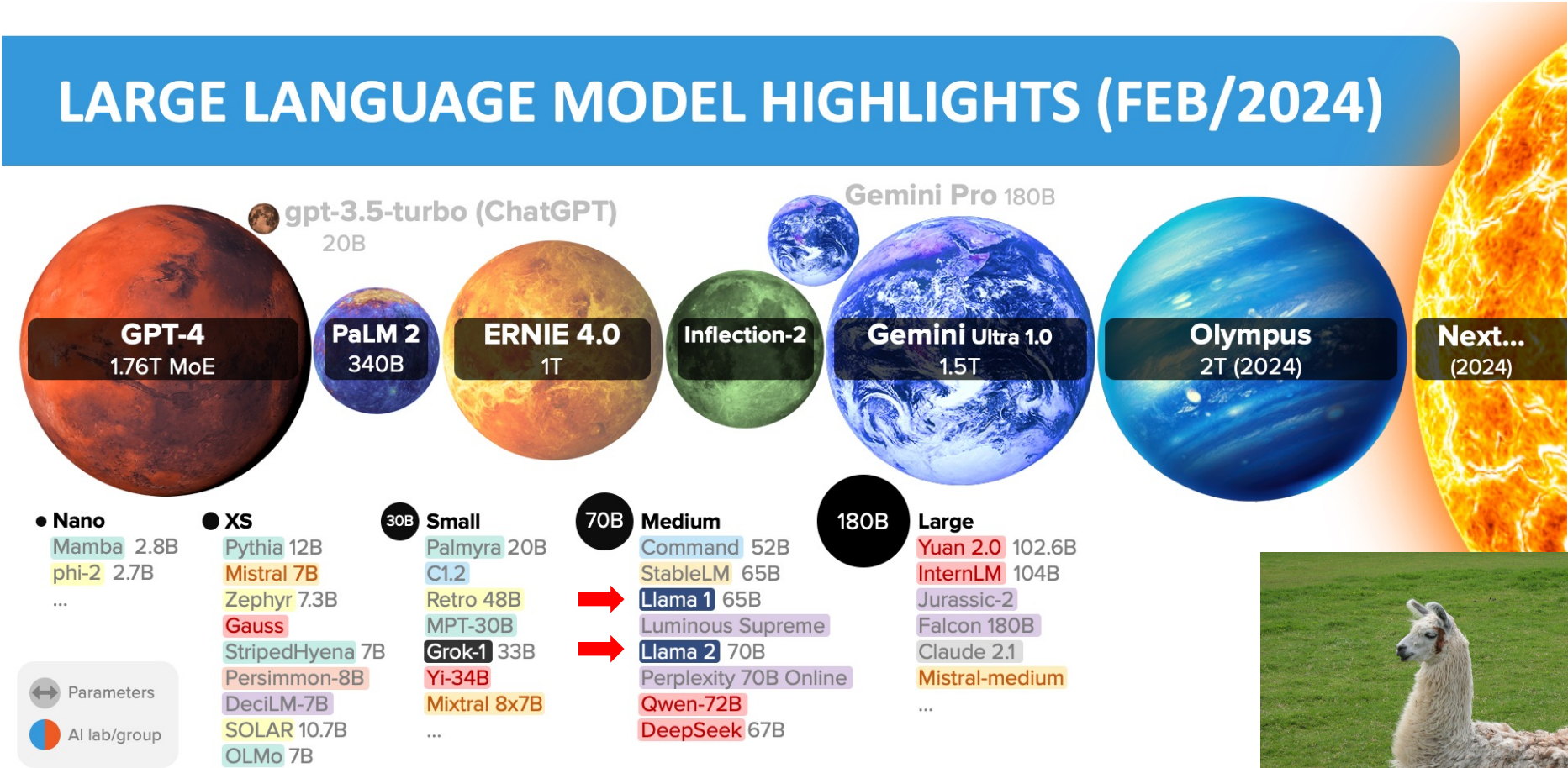


Figure source: <https://poole.ncsu.edu/thought-leadership/article/lets-chat-about-chatgpt/>
<https://openai.com/index/chatgpt/>

大型語言模型參數量非常多



Sizes linear to scale. Selected highlights only. All models are available. All models are Chinchilla-aligned (20:1 tokens:parameters) <https://lilearchitect.ai/chinchilla/> All 200+ models: <https://lilearchitect.ai/models-table/> Alan D. Thompson, 2023-2024.

Source: Inside language models (from GPT-4 to PaLM) – Dr Alan D. Thompson – Life Architect



GPU Memory Estimated (Model weights)

以 Llama 2-7B 為例，且儲存參數的精度為 float 32 (FP32)：

在 float 32 (FP32) 的情況下，1 個參數需要 4 bytes 儲存

-> 7B = 70億個參數 = $7,000,000,000 * 4 \text{ bytes}$ = (扣掉3個逗號從B換成GB) **28GB**

數值範圍比較

1 個 byte = 8 個 bits

數值類型名稱	PyTorch dtype	數值範圍	Bit數	Byte數
float32	torch.float32	約 $\pm 3.4e38$	32	4
float16	torch.float16	約 $\pm 6.5e4$	16	2
int8	torch.int8	-128 ~ 127	8	1
uint8	torch.uint8	0 ~ 255	8	1
int4	x	-8 ~ 7	4	0.5
uint4	x	0 ~ 15	4	0.5

u: unsigned (無號，代表數值只有正的)

Introduction to Quantization (量化)

(這頁只是示意圖，數值不精準)

$$\begin{bmatrix} -0.4 & 1.3 & 3.73 \\ -4.7 & -3.2 & -6.4 \\ 8.5 & 14.3 & 13.5 \end{bmatrix} \xrightarrow[\text{FP32} \rightarrow \text{int8}]{\text{降低儲存精度}} \begin{bmatrix} 0 & 1 & 4 \\ -5 & -3 & -6 \\ 9 & 14 & 14 \end{bmatrix}$$

36 bytes **8 bytes**

32-bit floating point (FP32): 1
個值需要**4個bytes**才能儲存

8-bit Integer (int8): 1個值需要
1個bytes才能儲存

誤差：

$$\begin{bmatrix} 0.4 & -0.3 & 0.27 \\ -0.3 & 0.2 & 0.4 \\ 0.5 & -0.3 & 0.5 \end{bmatrix}$$

量化後的模型非常常見

<https://huggingface.co/unsloth/Kimi-K2-Thinking-GGUF>

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Community](#) [Docs](#) [Enterprise](#) [Pricing](#)

 **unsloth/Kimi-K2-Thinking-GGUF** [like](#) 98 [Follow](#) [Unsloth AI](#) 10.8k

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[Model card](#) [Files and versions](#) [xet](#) [Community](#) 5

[Deploy](#) [Use this model](#)

[Edit model card](#)

Read our How to Run Kimi-K2 Guide!

Nov 8: We collabed with the Kimi team on a [system prompt fix](#).

Unsloth Dynamic 2.0 achieves superior accuracy & outperforms other leading quants.

 **unsloth** [Join our Discord](#) [Documentation](#)

It is recommended to have 247 GB of RAM to run the 1-bit Dynamic GGUF.

To run the model in full precision, you can use 'UD-Q4_K_XL', which requires 646 GB RAM.

Downloads last month
40,732



[GGUF](#) [Model size](#) [1T params](#) [Architecture](#) [deepseek2](#) [Chat template](#)

Hardware compatibility

Apple M3 (16 GB) x1

1-bit	IQ1_S 285 GB	TQ1_0 247 GB	IQ1_M 309 GB				
2-bit	IQ2_XXS 335 GB	Q2_K 374 GB	IQ2_M 354 GB	Q2_K_L 374 GB	Q2_K_XL 386 GB		
3-bit	IQ3_XXS 422 GB	Q3_K_S 443 GB	Q3_K_M 490 GB	Q3_K_XL 455 GB			
4-bit	IQ4_XS 547 GB	Q4_K_S 583 GB	IQ4_NL 579 GB	Q4_0 581 GB	Q4_1 643 GB	Q4_K_M 621 GB	Q4_K_XL 646 GB
5-bit	Q5_K_S 707 GB	Q5_K_M 729 GB	Q5_K_XL 730 GB				
6-bit	Q6_K 843 GB	Q6_K_XL 879 GB					
8-bit	Q8_0 1.09 TB	Q8_K_XL 1.19 TB					
16-bit	BF16 2.05 TB						

Kimi K2: Open Agentic
Intelligence

GPU Memory Estimated (Model weights)

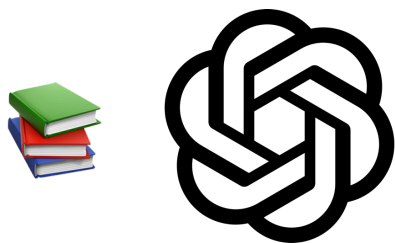
以 Llama 2-7B 為例，且儲存參數的精度為 float 32 (FP32)：

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-> 7B = 70億個參數 = $7,000,000,000 * 4 \text{ bytes}$ = (扣掉3個逗號從B換成GB) **28GB**

模型	參數量	Memory (FP32)	Memory (FP16)
DeepSeek v3	685B	2740 GB	1370 GB
Llama 4 Scout	109B	436 GB	218 GB
GPT-2 XL	1.5B	6 GB	3 GB

讓大型語言模型適用於你的任務？



翻譯、聊天、寫故事 ...



醫學資料、少數語言、網路上查不到的知識



解決方案：增加提示詞 或 訓練語言模型

Figure source:
https://gameofthrones.fandom.com/wiki/High_Valyrian?file=Nekesse_Valyrio.jpg

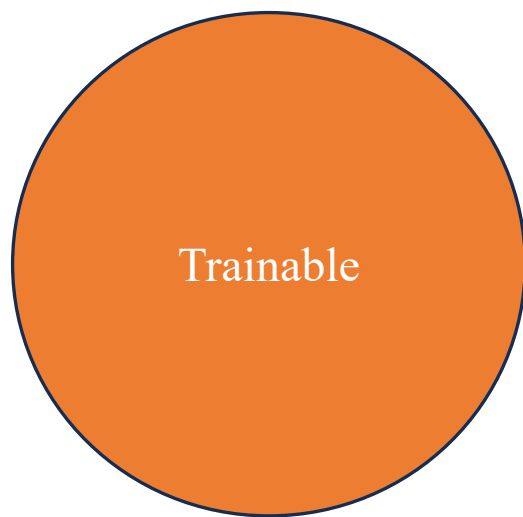
GPU Memory Estimated (Full Fine-tuning)

Llama 2-7B		
16-bit float, max_length (seq) = 4096, hidden_size = 4096, batch_size (bs) = 1		
	算法	Memory (概略)
CUDA	-	1 GB
Model weights	$\text{size}(\text{float}) * N_{\text{parameter}}$	14 GB
Gradients	$\text{size}(\text{float32}) * N_{\text{trainable}}$	28 GB
Hidden states	$\text{size}(\text{float}) * \text{seq} * \text{hidden_size} * L$	1 GB
Optimizer states (Adam)	$2 * \text{size}(\text{float32}) * N_{\text{trainable}}$	56 GB

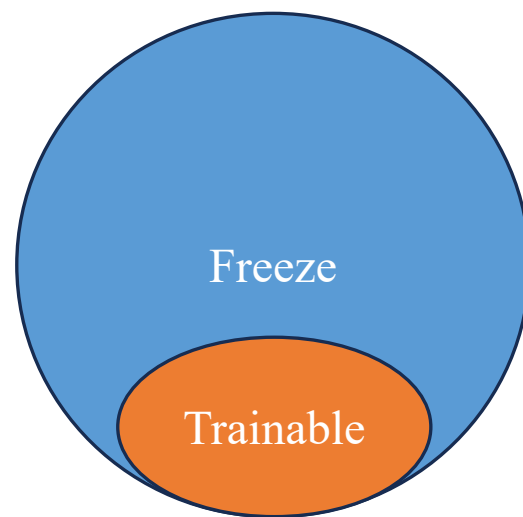
L : Number of layers in model (eq. 32 layers)

Estimate: ~100 GB
*NVIDIA 5090: 32GB

有沒有可能只微調 LLM 一部分的參數呢？



Full Fine-tuning
Update **all model parameters**



Parameter-efficient Fine-tuning (PEFT)
Update a **small subset** of model parameters

GPU Memory Estimated (PEFT)

- 假設我們只更新 2% 的參數

Llama 2-7B		
16-bit float, max_length (seq) = 4096, hidden_size = 4096, batch_size (bs) = 1		
	算法	Memory
CUDA	-	1 GB
Model weights	$\text{size(float)} * N_{\text{parameter}}$	14 GB
Gradients	$\text{size(float32)} * N_{\text{trainable}}$	$28 * 0.02 = 0.56 \text{ GB}$
Hidden states	$\sim \text{size(float)} * \text{seq} * \text{hidden_size} * L$	1 GB
Optimizer states (Adam)	$2 * \text{size(float32)} * N_{\text{trainable}}$	$56 * 0.02 = 1.12 \text{ GB}$

L : Number of layers in model (eq. 32 layers)

Estimate: 17.68 GB

*NVIDIA 3090: 24 GB

PEFT Outline

[Adapters](#)

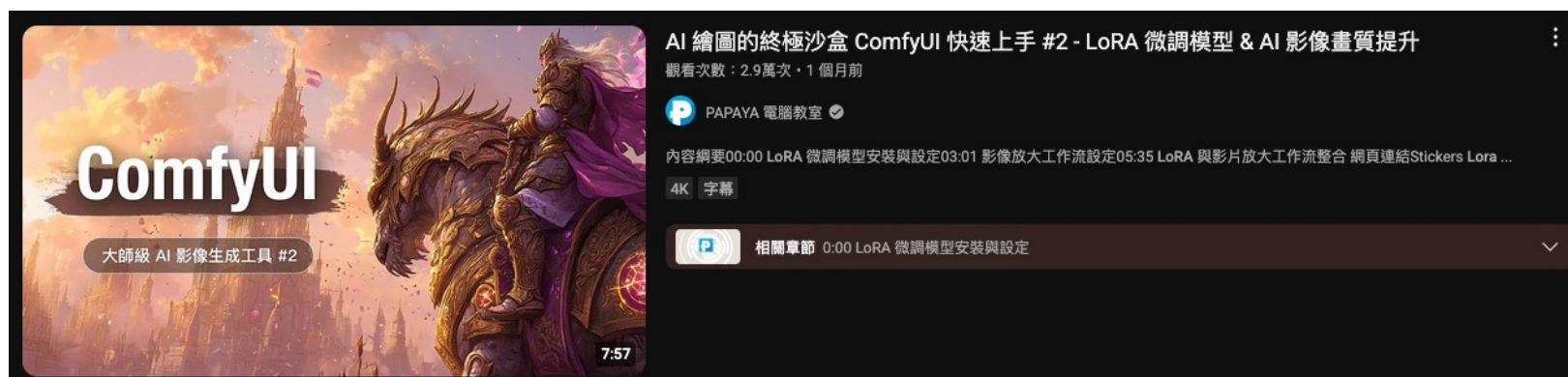
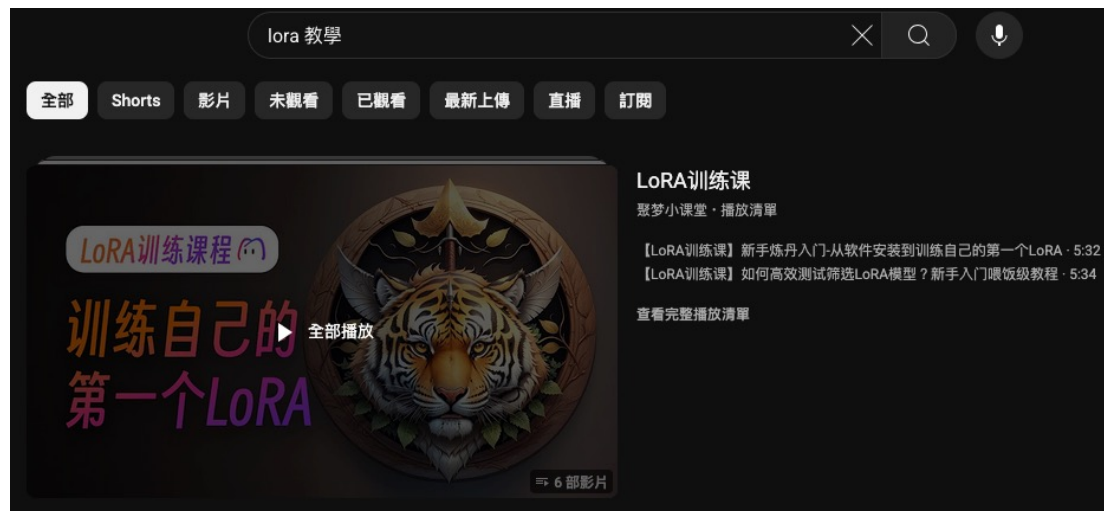
[LoRA](#)

[Prefix-Tuning](#)

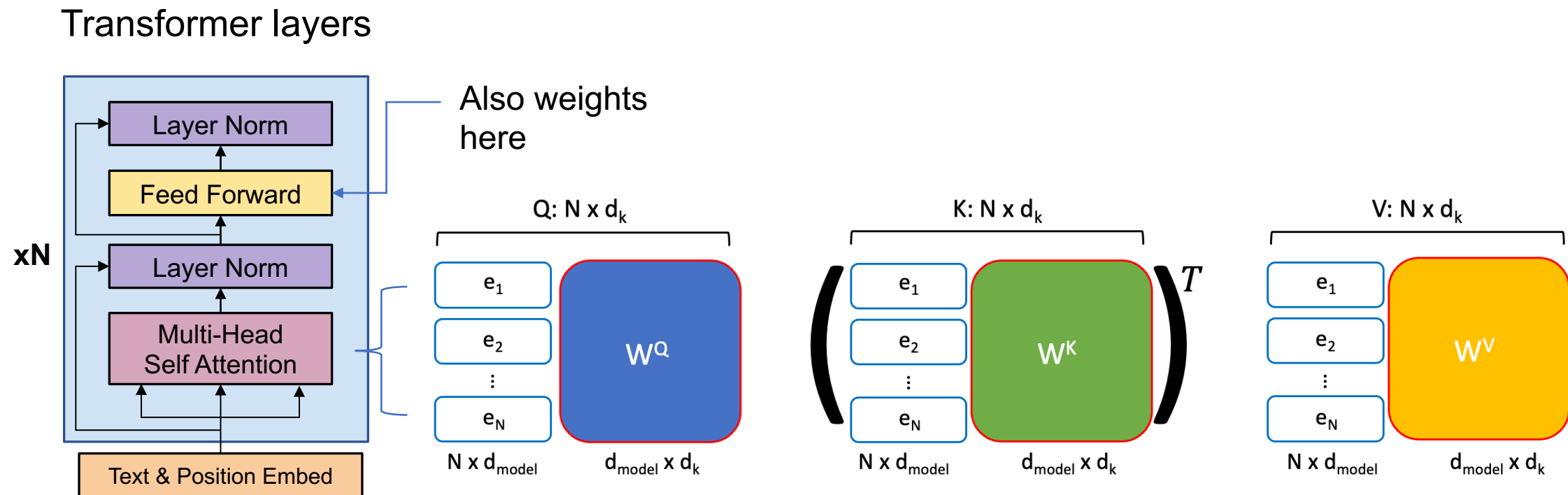
[BitFit](#)

LoRA: **Low-Rank A**daptation

LoRA is common ...

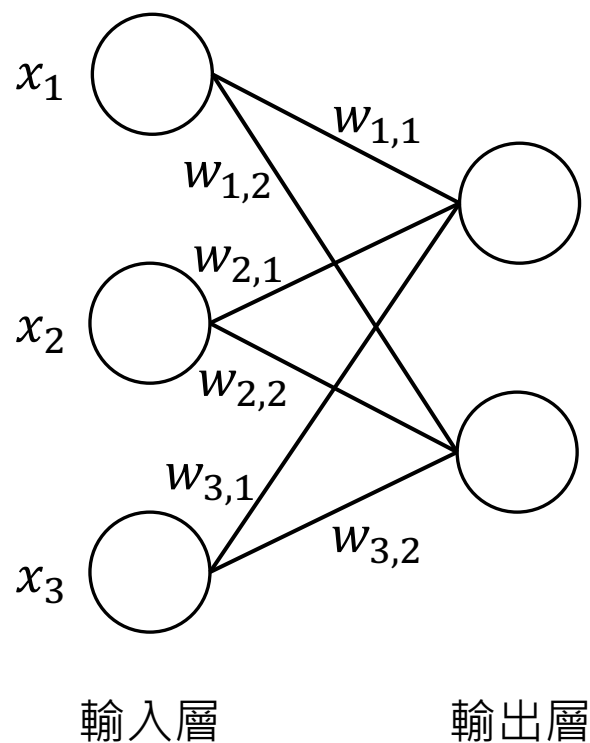


[Prerequisite] Weights in Transformer layers



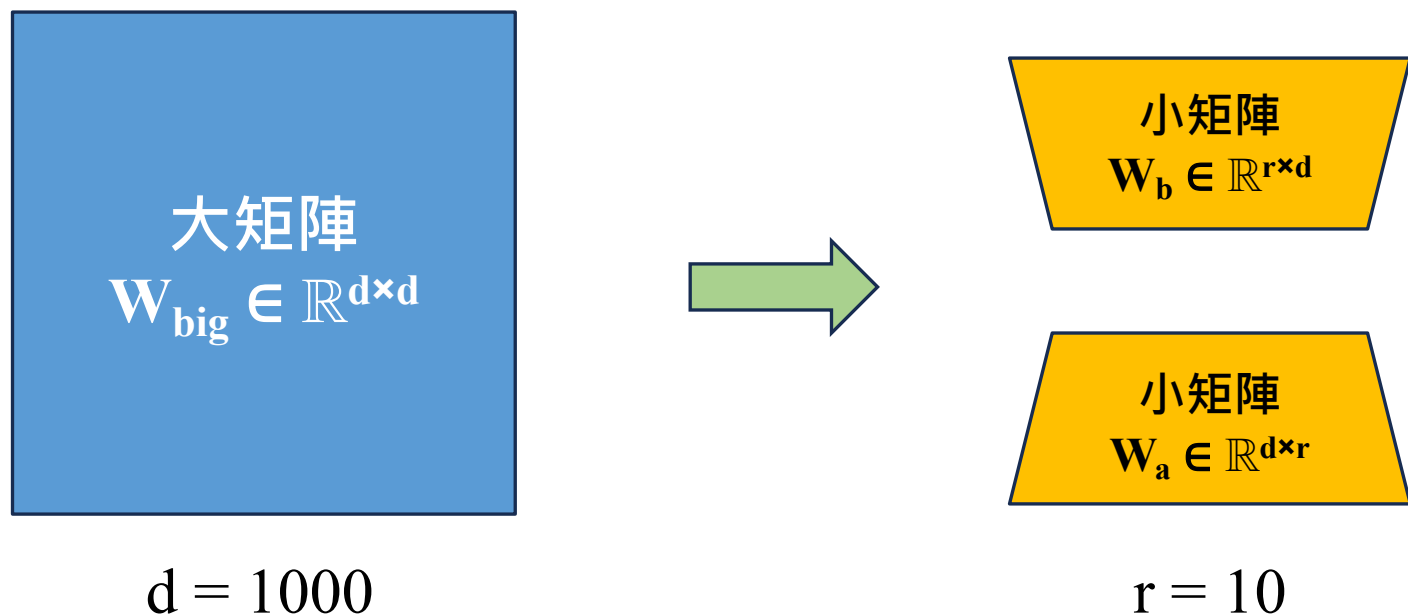
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

[Recap] MLP is composed of weight matrices



$$\begin{matrix} [x_1 & x_2 & x_3] \\ \mathbb{R}^{1 \times 3} \end{matrix} \times \begin{matrix} \boxed{\begin{bmatrix} w_{1,1} & w_{1,2} \\ w_{2,1} & w_{2,2} \\ w_{3,1} & w_{3,2} \end{bmatrix}} \\ \mathbb{R}^{3 \times 2} \end{matrix} + \text{bias} \quad \mathbb{R}^2$$

[Prerequisite] 矩陣分解



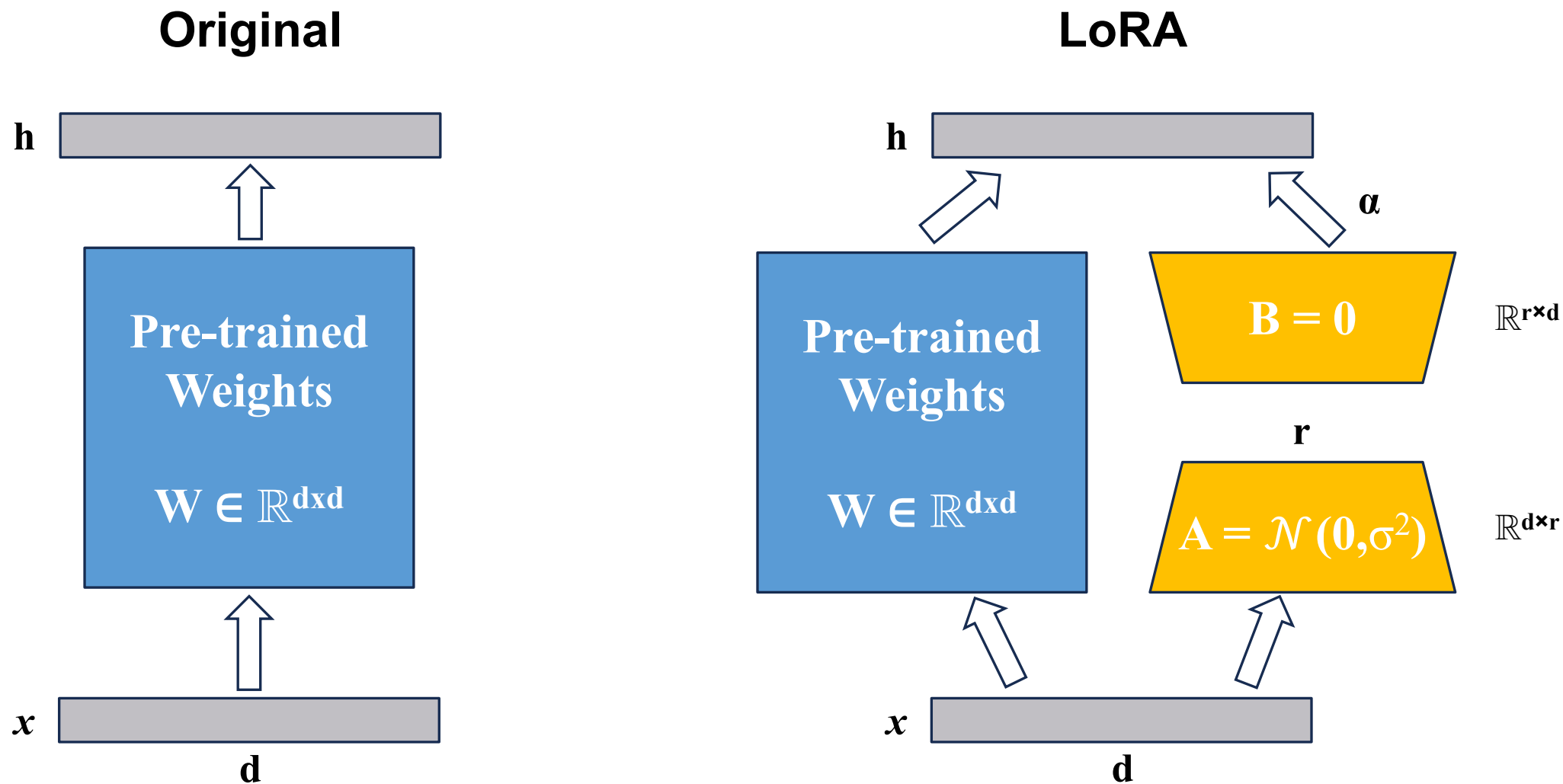
參數量

$$1000 \times 1000 = 1 \text{ 百萬}$$

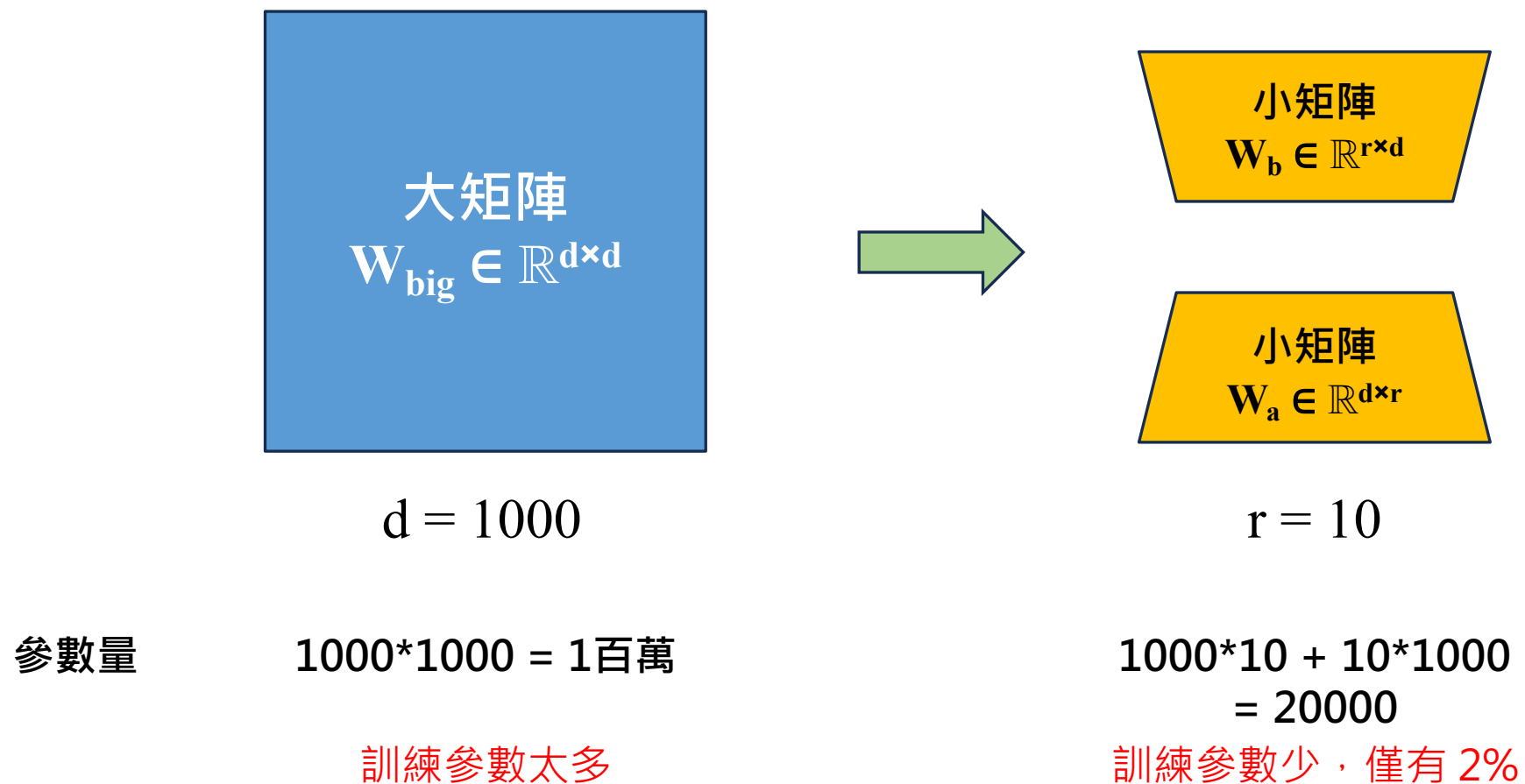
$$1000 \times 10 + 10 \times 1000 = 20000$$

LoRA: Low-Rank Adaptation

只有黃色部分會被訓練
 α 是超參數



為什麼 LoRA 要這樣做？



Low-Rank 的部分在哪？(1)

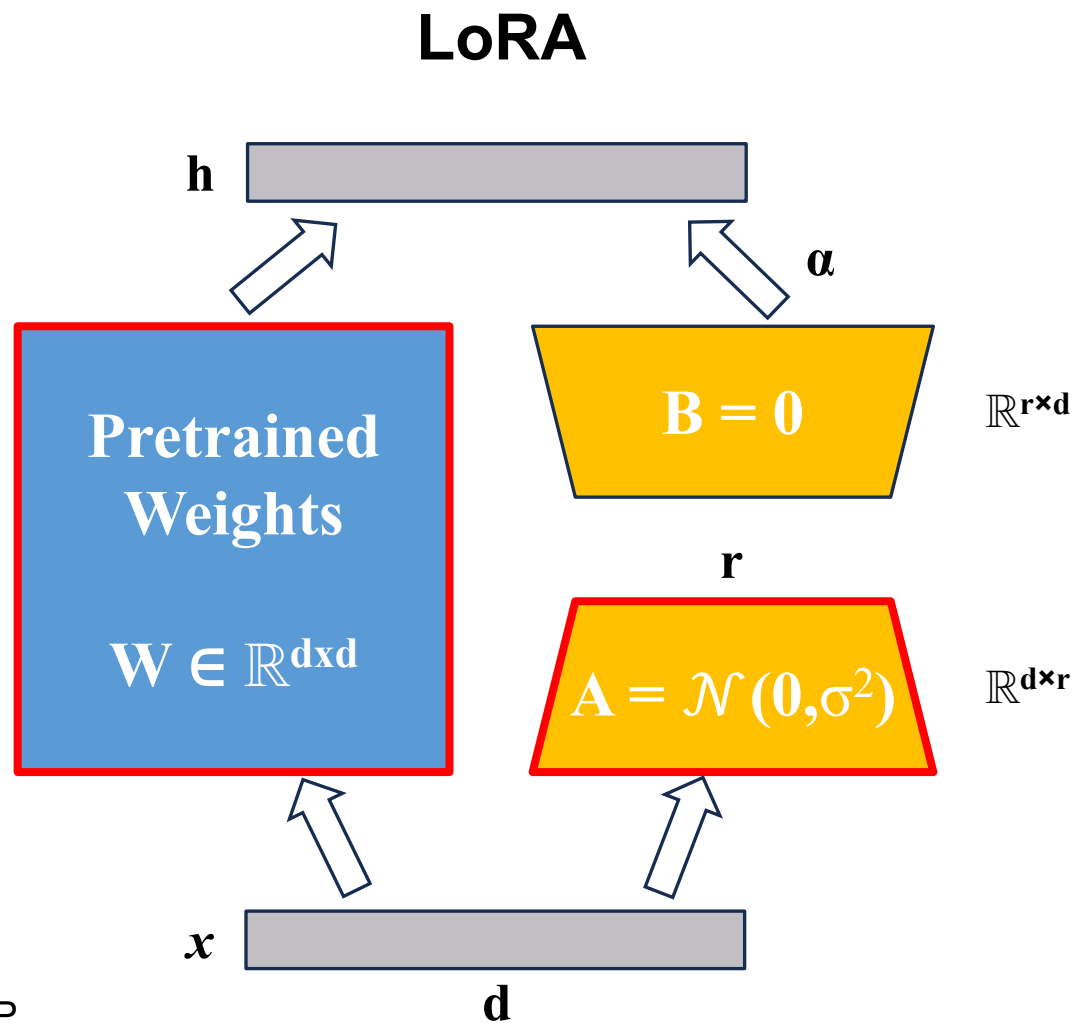
$$\begin{bmatrix} \checkmark & \checkmark & & \checkmark \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Reduced row-
echelon form

$\checkmark$: Linearly-independent vectors
rank = 3

rank 大小最多等於 column
vectors (或 row vectors) 的數量

Low-Rank 的部分在哪？(2)



$d > r$ 的情況下 (通常 $d \gg r$)，**A** 或 **B** 的 **rank** 通常遠小於 **W** 的 **rank** 故為 low-rank 的由來

LoRA: **Low-Rank** **A**daptation (w / pseudo code)

Pseudocode:

```
input_dim = 768 # the hidden size of the pre-trained model
output_dim = 768 # the output size of the layer
rank = 8 # The rank 'r' for the low-rank adaptation
```

```
W = ... # from pretrained network with shape input_dim x output_dim
```

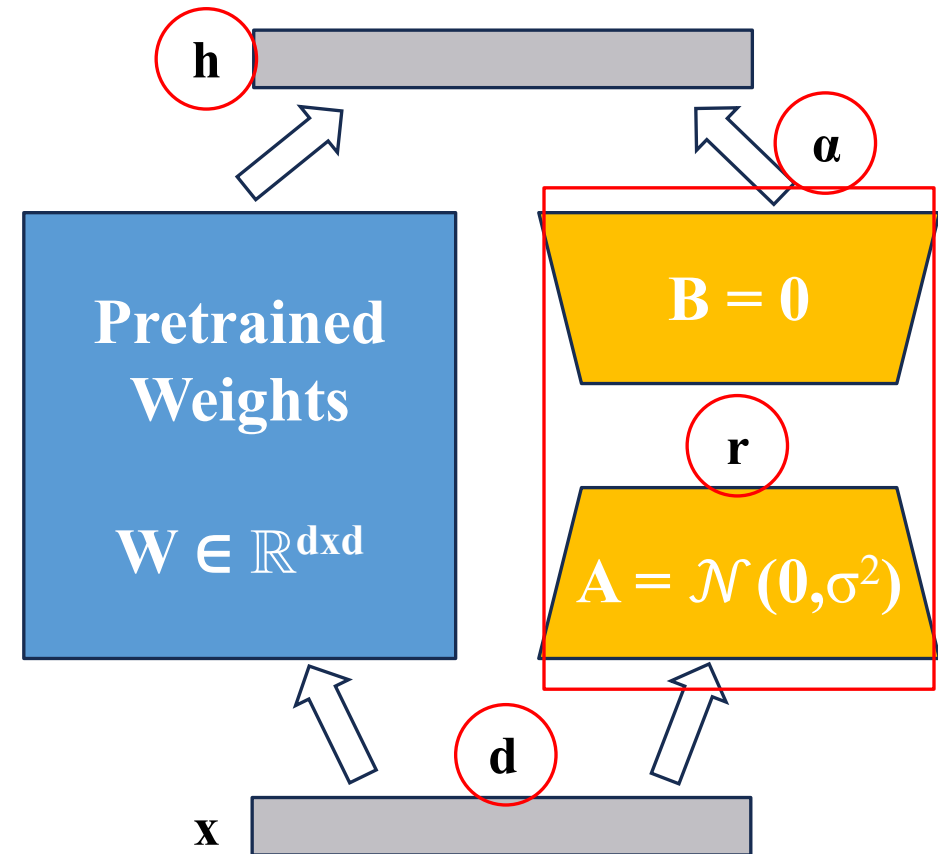
```
W_A = nn.Parameter(torch.empty(input_dim, rank)) # LoRA weight A
W_B = nn.Parameter(torch.empty(rank, output_dim)) # LoRA weight B
```

```
# Initialization of LoRA weights
```

```
nn.init.kaiming_uniform_(W_A, a=math.sqrt(5))
nn.init.zeros_(W_B)
```

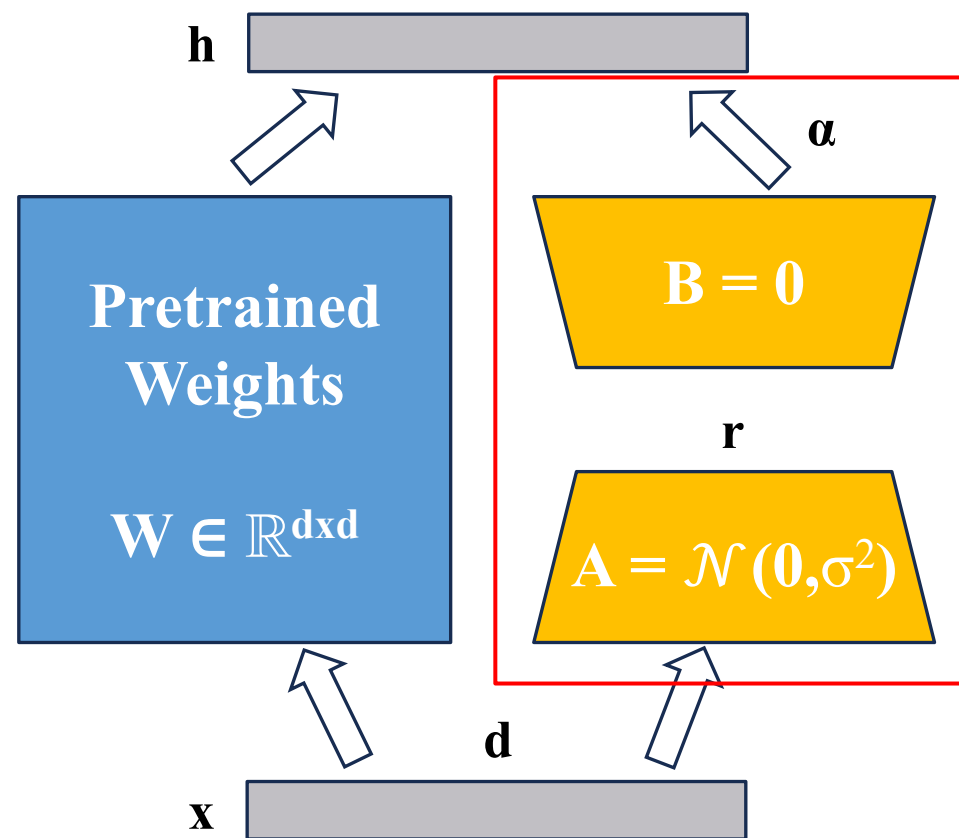
```
def regular_forward_matmul(x, W):
    h = x @ W
    return h
```

```
def lora_forward_matmul(x, W, W_A, W_B):
    h = x @ W # regular matrix multiplication
    h += x @ (W_A @ W_B) * alpha # use scaled LoRA weights
    return h
```



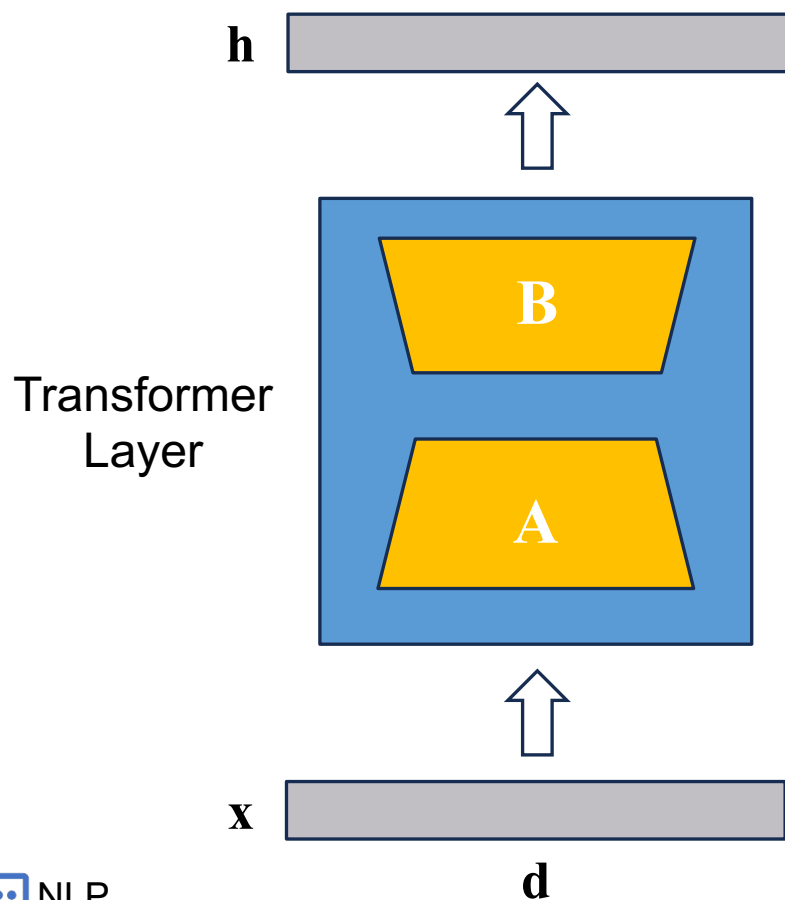
[注意事項] LoRA 的問題

- r 的數值大小需要手動調整
 - r 越小，訓練參數量越少
- 相較於原本沒有 LoRA 的模型，
LoRA 其實會讓 inference 速度變慢



可不可以不要外加？(Adapter 作法)

Adapter: Parameter-Efficient Transfer Learning for NLP. ICML 2019



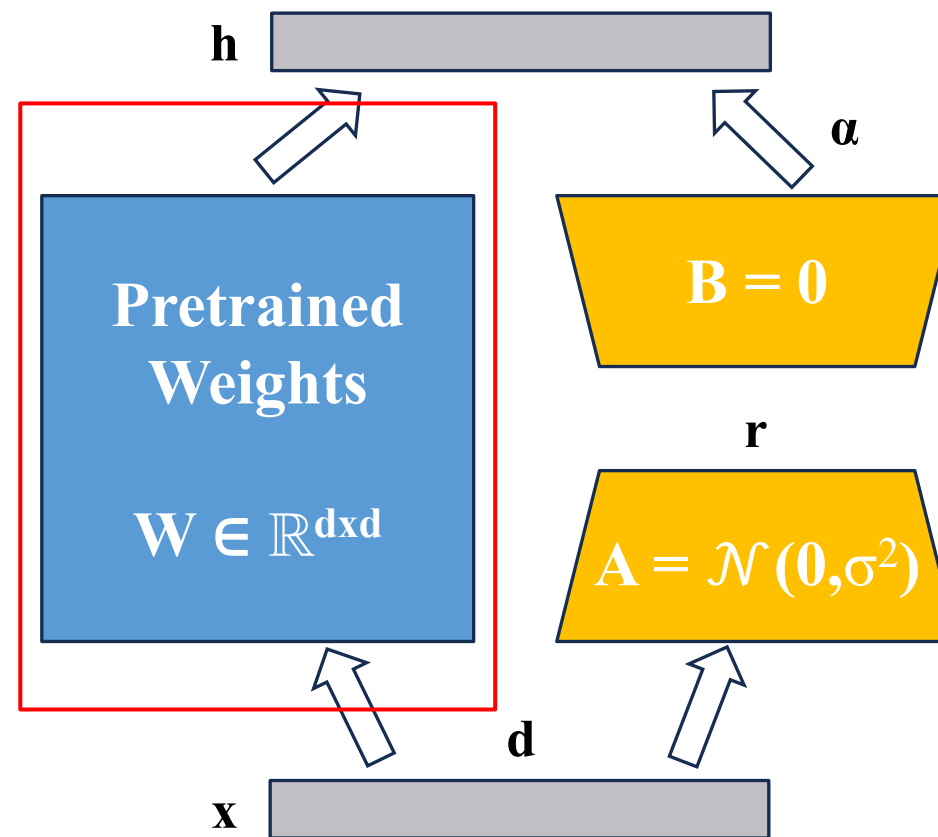
優點：模型設計更靈活

缺點：

- 直接變更了模型架構 (串聯)
 - -> 計算時間比 LoRA 長

[注意事項] LoRA 的問題

- r 的數值大小需要手動調整
 - r 越小，訓練參數量越少
- 相較於原本沒有 LoRA 的模型，LoRA 其實會讓 inference 速度變慢
- 訓練和 inference 時，大模型的部分其實還是被載入記憶體...



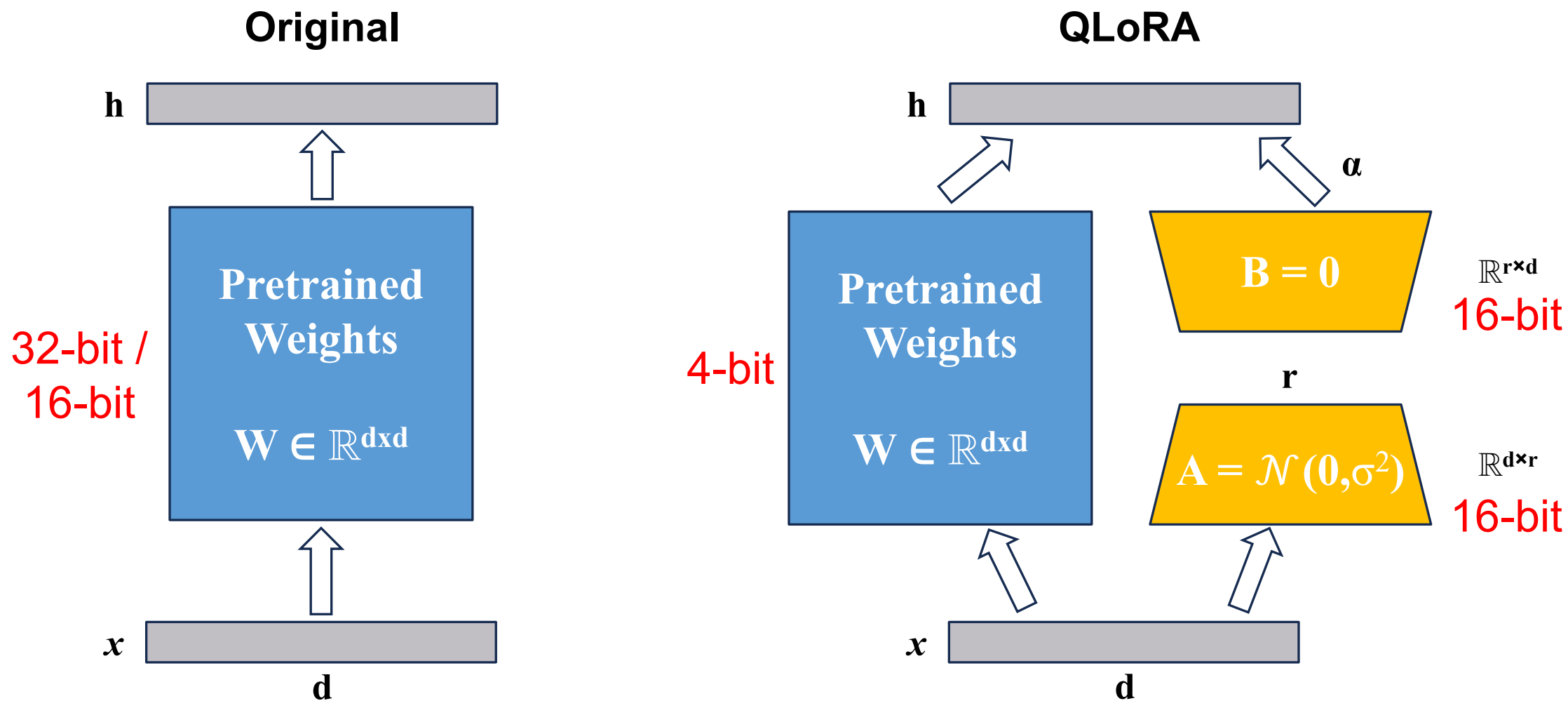
LoRA 可能可以在少量資料下進行微調

Method	MNLI(m)-100	MNLI(m)-1k	MNLI(m)-10k	MNLI(m)-392K
GPT-3 (Fine-Tune)	60.2	85.8	88.9	89.5
GPT-3 (PrefixEmbed)	37.6	75.2	79.5	88.6
GPT-3 (PrefixLayer)	48.3	82.5	85.9	89.6
GPT-3 (LoRA)	63.8	85.6	89.2	91.7

Table 16: Validation accuracy of different methods on subsets of MNLI using GPT-3 175B. MNLI- n describes a subset with n training examples. We evaluate with the full validation set. LoRA performs exhibits favorable sample-efficiency compared to other methods, including fine-tuning.

QLoRA: **Quantized** LoRA

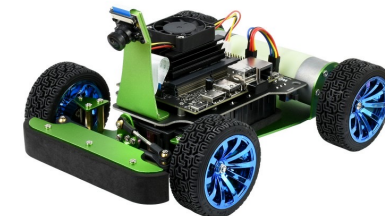
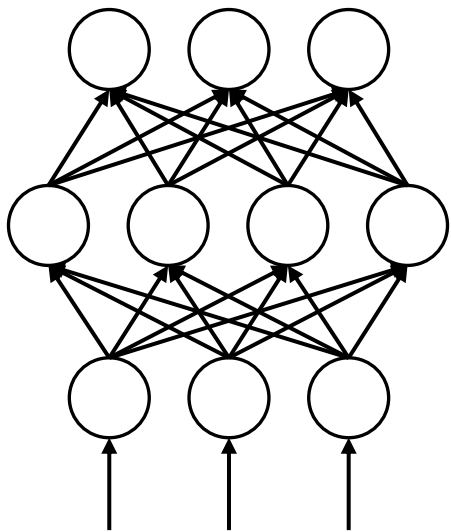
只有黃色部分會被訓練
 α 是超參數



[注意事項] QLoRA

- 記憶體使用量： $\text{QLoRA} < \text{LoRA}$
- 模型效能： $\text{QLoRA} \leq \text{LoRA}$

How about model compression?



Comparison: What is Model Compression?



Original Model



Quantized Model



Pruned Model



Distilled Model

PEFT vs. Model compression

	PEFT	Model Compression
目標	讓模型適應新的任務， 但模型大小不變	加速模型運算或模型儲存空間
相較於原始模型的 改變內容	插入少量可訓練參數	減少整體模型結構或權重
參數更新	只更新少量新參數	先減少整體模型結構後針對新的模型進行訓練
使用情境	需要模型學會新的任務時	手機、邊緣裝置

Previous Videos of Model Compression

- https://youtu.be/Qqyo9U2JRBQ?si=V9F_nNm44dijmYKc
- <https://youtu.be/luBa01XYoBc?si=8Hj8fUUI6RwPPI1j>
- <https://youtu.be/dpE6LyIGU9w?si=5XBdr-JDJaHojoJD>

Announcement

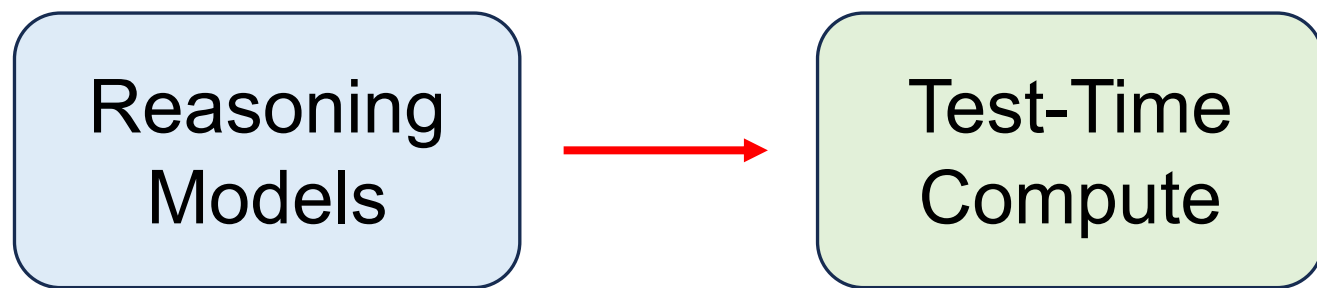
For Week 15
and Week 16

Week 15/16 報告順序與注意事項

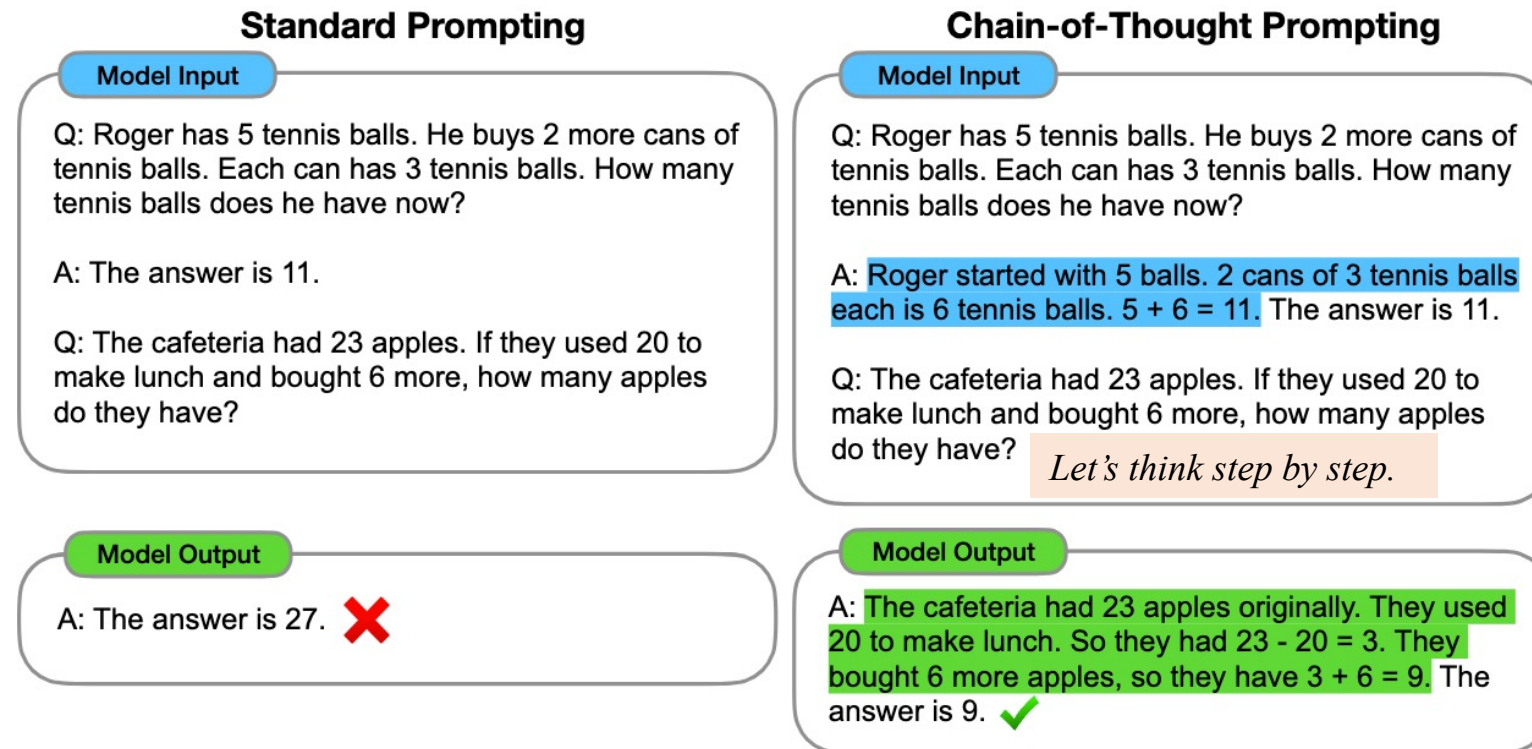
	A	B	C	D
1	組別編號	組長	人數	題目
2	3	蔡智翔	4	MAP - Charting Student Math Misunderstandings
3	8	陳威誠	3	MAP - Charting Student Math Misunderstandings
4	6	邱庭俞	4	MAP - Charting Student Math Misunderstandings
5	4	劉韋彤	5	NBME - Score Clinical Patient Notes
6	14	梁釗豪	4	NBME - Score Clinical Patient Notes
7	13	方翊丞	3	LLM Classification Finetuning
8	15	卓湧恩	3	Learning Agency Lab - Automated Essay Scoring 2.0
9	7	周芷妤	4	Learning Agency Lab - Automated Essay Scoring 2.0
10	以下為Week16報告組別			
11	2	胡樂麒	4	Learning Agency Lab - Automated Essay Scoring 2.0
12	1	巫煊崑	1	Learning Agency Lab - Automated Essay Scoring 2.0
13	12	汪朝臻	3	LLM Classification Finetuning
14	16	郭家瑋	4	LLM Classification Finetuning
15	10	鄭茹云	5	NBME - Score Clinical Patient Notes
16	5	陳彥佑	3	NBME - Score Clinical Patient Notes
17	9	譚曉雯	1	MAP - Charting Student Math Misunderstandings
18	11	謝易澄	4	MAP - Charting Student Math Misunderstandings

- 四個主題
- 報告順序可換
- 非報告組同學仍需到場
 - 聆聽報告
 - 進行同儕互評

Reasoning Models (會推理的模型)



Standard Prompting vs. Chain-of-Thought (CoT)



CoT:

<https://arxiv.org/abs/2201.11903>

Zero-shot-CoT:

<https://arxiv.org/abs/2205.11916>

Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted.

CoT - method

- Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022. By Google.

Prompt: Few-shot examples (Rationales written by human)

Question: Sammy wanted to go to where the people were. Where might he go?
Answer Choices: (a) populated areas (b) race track (c) desert (d) apartment (e) roadblock

Answer: The answer must be a place with a lot of people. Of the above choices, only populated areas have a lot of people. So the answer is (a) populated areas.

xN

Similar
in task

+

A gentleman is carrying equipment for golf, what is he likely to have?
Answer Choices: (a) populated areas (b) race track (c) desert (d) apartment (e) roadblock

(有選項的QA問題)



(模型回答)

Answer: (a) club. The answer must be something that is used for golf. Of the above choices, only clubs are used for golf. So the answer is (a) club.

Rationale
Answer

CoT is not just Few-shot -> [Recap] GPT-3

- The three settings explored for **in-context learning** in the GPT-3 paper:

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

- Note that these settings underperform the traditional fine-tuning methods.**

不同任務需要不同的 Prompt

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3 , which is less than water. Thus, a pear would float. So the answer is no.

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943. So the answer is 05/23/1943.

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer. So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya". So the answer is ya.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up. So the answer is no.

Zero-shot CoT

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Zero-shot CoT - method

1st prompting Reasoning Extraction

雞兔同籠，共有 35 個頭、94 隻腳。請問雞有幾隻？
Let's think step by step.



LLM



1. 假設 35 隻全部都是雞，則腳數會是 $35 \times 2 = 70$ 隻。
2. 但實際上有 94 隻腳，多出 $94 - 70 = 24$ 隻腳。
3. 每把一隻雞換成兔子，腳數會多 2 隻。
4. 因此兔子的數量是 $24 \div 2 = 12$ 隻。
5. 雞的數量是 $35 - 12 = 23$ 隻。

2nd prompting Answer Extraction

雞兔同籠，共有 35 個頭、94 隻腳。請問雞有幾隻？
Let's think step by step.

1. 假設 35 隻全部都是雞，則腳數會是 $35 \times 2 = 70$ 隻。
2. 但實際上有 94 隻腳，多出 $94 - 70 = 24$ 隻腳。
3. 每把一隻雞換成兔子，腳數會多 2 隻。
4. 因此兔子的數量是 $24 \div 2 = 12$ 隻。
5. 雞的數量是 $35 - 12 = 23$ 隻。

所以答案是 (阿拉伯數字)：



LLM



23

Zero-shot CoT 效果不差

	MultiArith	GSM8K
Zero-Shot	17.7	10.4
Few-Shot (2 samples)	33.7	15.6
Few-Shot (8 samples)	33.8	15.6
Zero-Shot-CoT	78.7	40.7
Few-Shot-CoT (2 samples)	84.8	41.3
Few-Shot-CoT (4 samples : First) (*1)	89.2	-
Few-Shot-CoT (4 samples : Second) (*1)	90.5	-
Few-Shot-CoT (8 samples)	93.0	48.7
Zero-Plus-Few-Shot-CoT (8 samples) (*2)	92.8	51.5
Finetuned GPT-3 175B [Wei et al., 2022]	-	33
Finetuned GPT-3 175B + verifier [Wei et al., 2022]	-	55

- Zero-shot CoT 仍輸給一般 CoT
- Let's think step by step 是個厲害的咒語

Let's think step by step on MultiArith

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	78.7
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		AbraKadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

可不可以讓模型自己長出推理能力？

Guo, Daya, et al. "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning." arXiv preprint arXiv:2501.12948 (2025).

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>...</think>` and `<answer>...</answer>` tags, respectively, i.e., `<think> reasoning process here </think>` `<answer> answer here </answer>`. User: **prompt**. Assistant:

用戶與助手之間的對話。

用戶提出問題，助手解答。助手先在腦海中推理，然後將答案提供給使用者。

推理過程和答案分別包含在 ``<think>...</think>`` 和 ``<answer>...</answer>`` 標籤內，即：

``<think>`此處為推理過程`</think>``，``<answer>`此處為答案`</answer>``。

使用者：**提示詞**。助手：

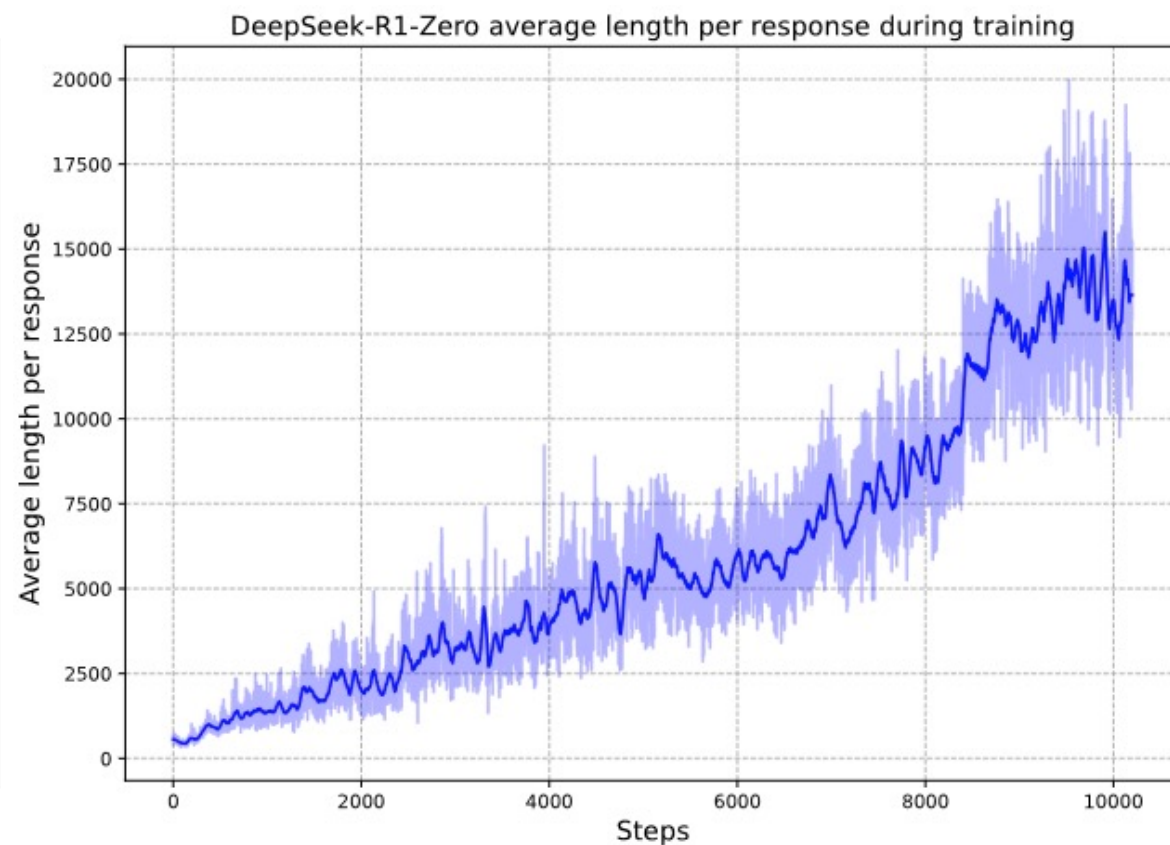
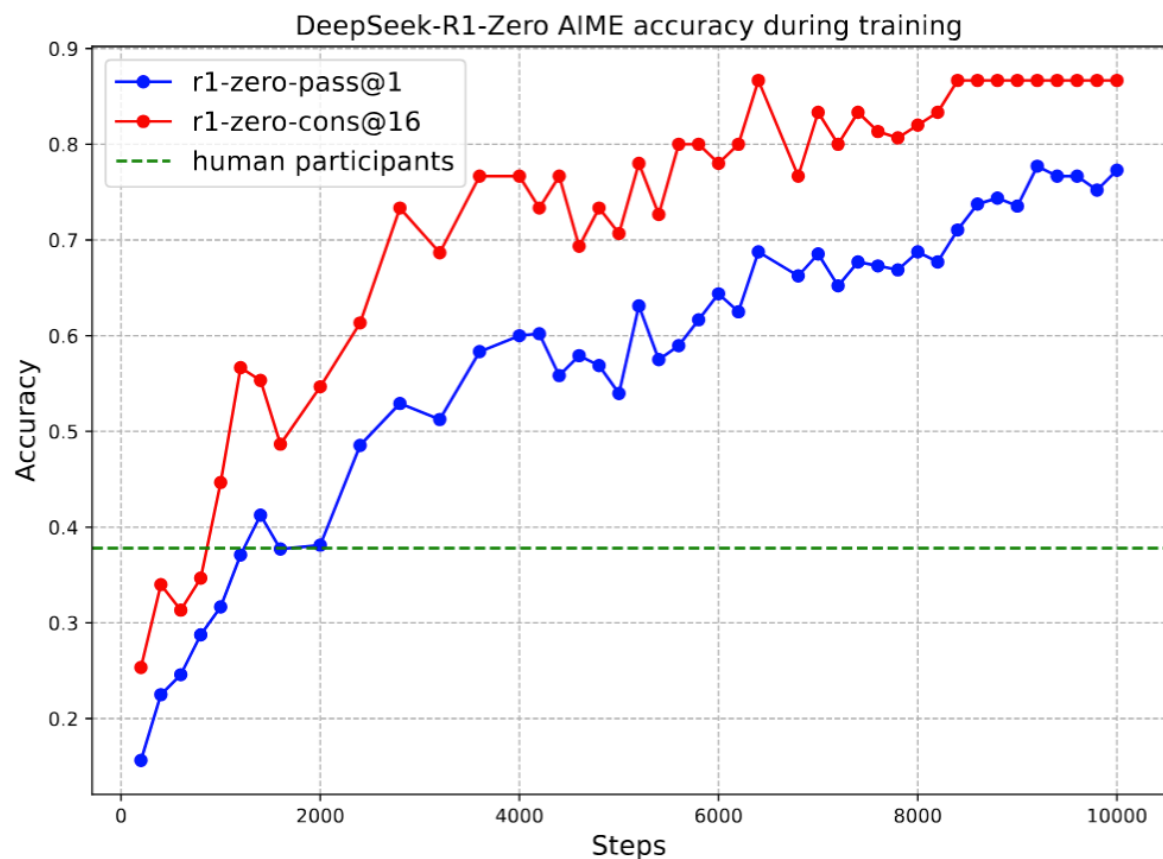
DeepSeek-R1-zero 細節

Guo, Daya, et al. "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning." arXiv preprint arXiv:2501.12948 (2025).

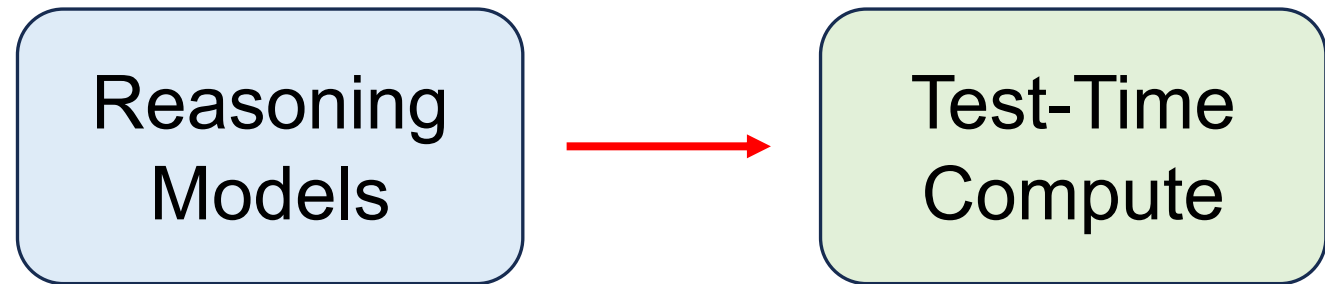
- 使用 RL 進行訓練 ([GRPO](#)，類似於RLHF的 [PPO](#))
- Two reward models (不需訓練):
 - **Answer rewards:** 答案是否正確
 - **Format rewards:** 有沒有把推理過程寫在 `<think>` 和 `</think>` 中
- 從 DeepSeek-V3-Base 作為初始化模型 (671B)
- 資料集：數學、程式設計、以及邏輯推論

DeepSeek-R1-zero 訓練過程

Guo, Daya, et al. "Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning." arXiv preprint arXiv:2501.12948 (2025).



Test-Time Compute



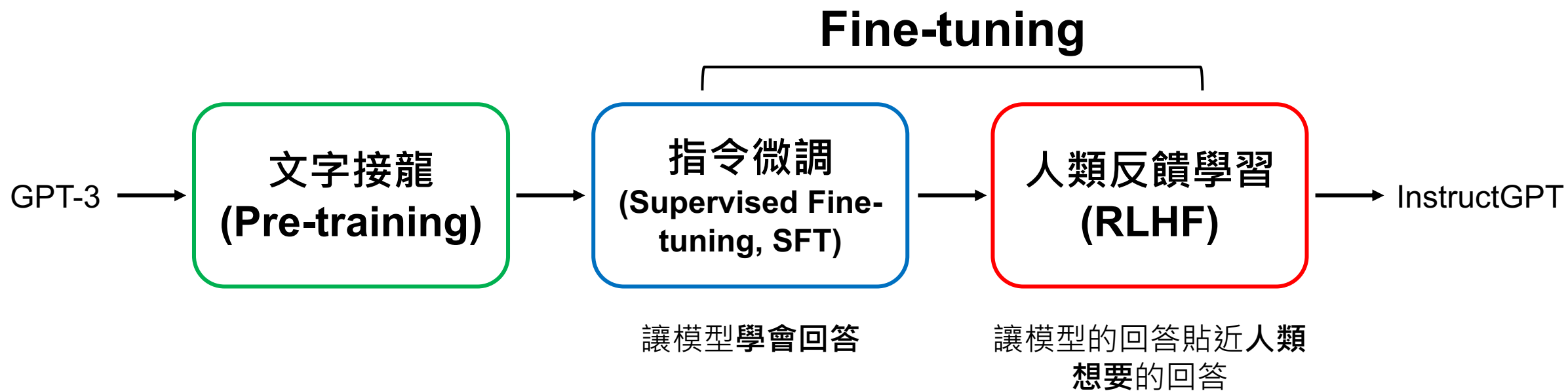
**Test time
Compute**



**One shot
Inference**

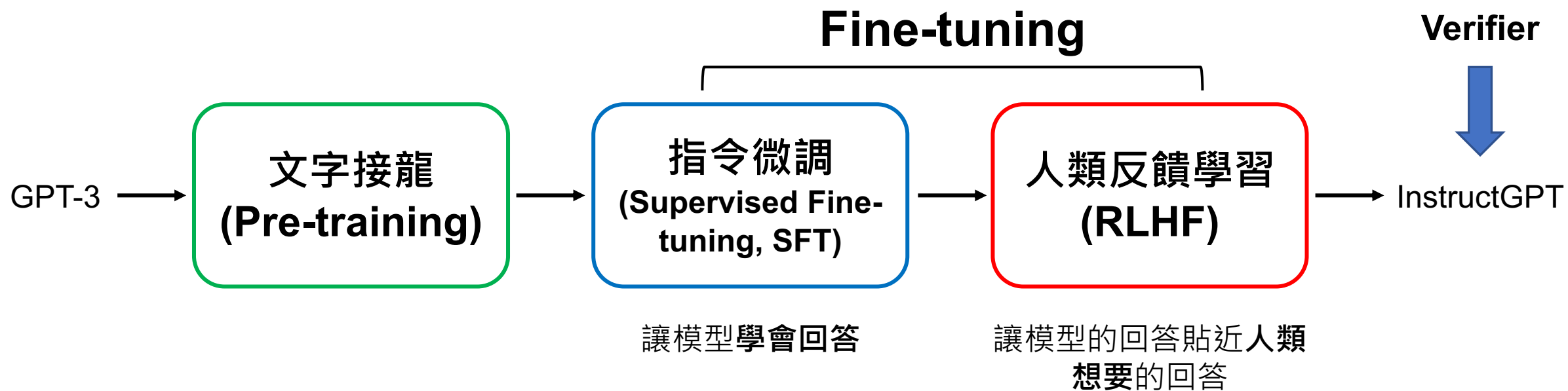
Figure ref: <https://medium.com/@gupta.moulik/is-this-something-that-will-take-us-to-agi-ttc-and-cot-explained-971fdf35f85c>

[Recap] The Pre-training then Fine-tuning Paradigm



RLHF: Reinforcement Learning with Human Feedback

增加驗證推理步驟的 Reward Model



PRM800K Dataset

Let's Verify Step by Step:
<https://arxiv.org/abs/2305.20050>

800K step-level labels across 75K solutions to 12K math problems

The denominator of a fraction is 7 less than 3 times the numerator. If the fraction is equivalent to $\frac{2}{5}$, what is the numerator of the fraction? (Answer: 14)

   Let's call the numerator x .

   So the denominator is $3x-7$.

   We know that $x/(3x-7) = 2/5$.

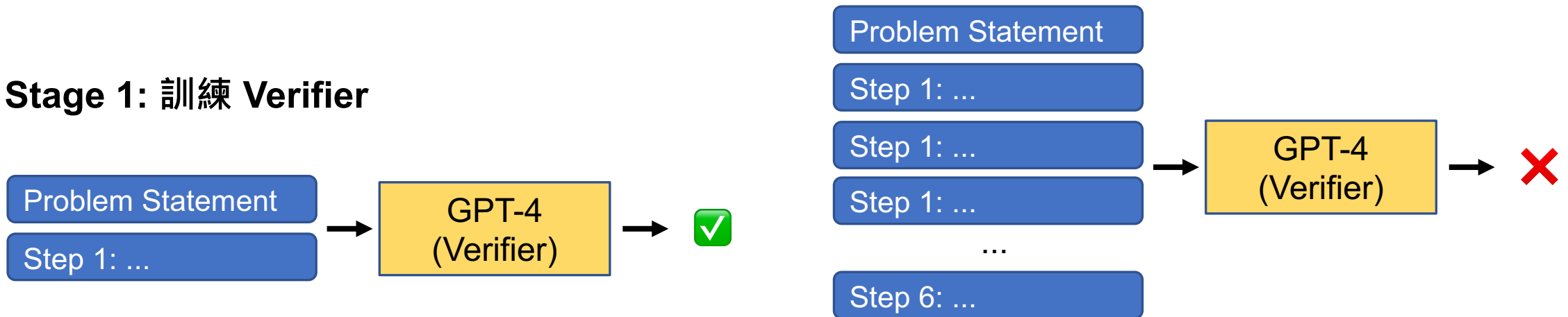
   So $5x = 2(3x-7)$.

   $5x = 6x - 14$.

   So $x = 7$.

Test-time Compute 步驟 (Stage 2)

Stage 1: 訓練 Verifier

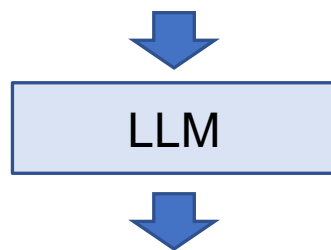


Let's Verify Step by Step:
<https://arxiv.org/abs/2305.20050>

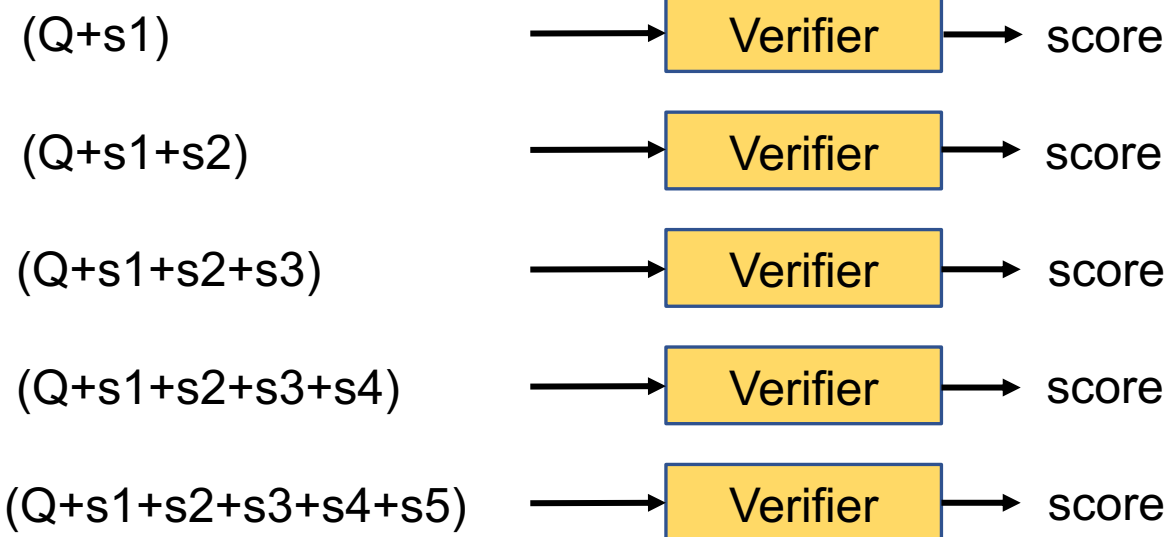
Test-time Compute 步驟 (Stage 2)

Stage 2: 進行 Test-time compute

Q: 雞兔同籠，共有 35 個頭、94 隻腳。請問雞有幾隻？
Let's think step by step.



- s1. 假設 35 隻全部都是雞，則腳數會是 $35 \times 2 = 70$ 隻。
- s2. 但實際上有 94 隻腳，多出 $94 - 70 = 24$ 隻腳。
- s3. 每把一隻雞換成兔子，腳數會多 2 隻。
- s4. 因此兔子的數量是 $24 \div 2 = 8$ 隻。
- s5. 雞的數量是 $35 - 8 = 27$ 隻。



Thank you!

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